MIL-M-52320C
8 October 1968
SUPERSEDING
MIL-M-0052329B(ME)
21 June 1967
MIL-M-52329A
1 April 1965

MILITARY SPECIFICATION

MIXING PLANT, ASPHALT, CONTINUOUS-MIX, DIESEL-ENGINE-DRIVEN, PORTABLE, 100-150 TONS PER HOUR CAPACITY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a high-type, continuous-mix, portable, asphalt mixing plant. The plant is comprised of towable vehicles plus components and accessories which can be transported by truck or trailer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	Bearing, Roller, Tapered.
HH-B-671	Bricks , Refractory: Fire Clay:
HH-R-191	Refractories; Fire-Clay, Plastic.
QQ-S-781	Steel, Strapping, Flat.
QQ-S-790	Steel Strapping, Round (Bare and Zinc-Coated).
TT-P-664	Primer Coatings, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	- Varnish, #par, Water-Resisting.
UU-T-81	Tags, Shipping and Stock.
VV-F-815	- Fuel Oil, Burner.

FSC 3895

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WW-P-406	- Pipe, Steel (Seamless and Welded) (for Ordinary Use).
WW-U-531	- Unions, Pipe Steel or Malleable Iron; Threaded Connection.
WW-V-54	- Valve, Gate, Bronze, 125- and 159-Pound, Screwed and Flanged (for Land Use).
ZZ-B-206	- Belting, Flat; Conveyor, Rubber.
KKK-L-370	- Lining, Friction (Clutch and Brake; Metallic, Metal-Ceramic and Semimetallic).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-3-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Pressure-sensitive Adhesive, Waterproof, for Packaging.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-C-392	- Conveyor, Belt: Portable.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-V-1174	- Varnish, Spar, Water Resisting (Fomula No. 80).
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine (Heavy Duty).
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, wood; Open, 12,000- and 16,000-Pound capacity.
MIL-M-3971/1-1	- Meters, Time Totalizing, Non-Hermetically Sealed, Electrical; Type I, 4-40 V or 40-130 V; Grades A and B, 2-1/2 Inch and 3-1/2 Inch, 3 Hole Flange.
MIL-H-3992	- Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive.

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MIL-S-8660 - Silicone Compound.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-E-11275 - Engines, Gasoline, Industrial Type, General Specification for.
MIL-E-11276 - Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-C-13486/12 - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Multiple Conductors Unshielded.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.
MIL-P-13983 - Paints Temporary, Lusterless, Gasoline Removable.
MIL-P-14105 - Paint, Heat-Resisting (for Steel Surfaces).
MIL-I-15126 - Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-H-15424 - Hand Tools; Packaging of.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Bread-In.
MIL-L-22851 - Lubricating Oil, Aircraft Piston Engine (Ashless Dispersant).
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-P-46093 - Primer Coating, Synthetic (for Brake Drum).
MIL-L-45199 - Lubricating Oil, Internal Combustion Engine (High Output Diesel).
MIL-T-46786 - Tanks, Fuel, Engine, General Requirements for.
MIL-A-52363 - Air Clearners, Intake; Dry-Type (for Internal Combustion Engine).

STANDARDS

Federal

FED. STD. No. 595 - Colors.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

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MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-421	- Chain, Roller; Power Transmission and Conveyors, Flat Link Plates; Base Pitch, Single and Multiple Strand; Connecting Links and Attachment Links.
MIL-STD-423	- Chain, Roller; Conveyor, Standard and Large Rollers, Flat Link Plates, Double Pitch, Single Strand; Connecting Links and Attachment Links.
MIL-STD-424	- Chain, Roller; Power Transmission, Offset Sidebars, Single-Pitch.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1224	- Visual Inspection Guide for Pneumatic Tires (Nonaircraft). -
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MS16961	- Troughing Carrier Assembly, 4 In. Dia Roll, 4-5/8 In. Backing; 24 and 30 In. Belt Width.
MS16964	- Return Roll Assembly, 4 In. Dia Roll, for 4-5/8 In. Backing Trough Carrier; 24 and 30 In. Belt Width.
MS17160	- Clip, Spring Tension (for Cylindrical Electronic Component).
Ms21322	- Clamps, Loop - Aluminum - Plain and Cushioned.
MS21333	- Clamps, Loop - Steel - Plain and Cushioned.
MS21334	- Clamps, Loop - Double, Cable or Tube, Steel, 'Plain and Cushioned.
MS27142	- Connector, Plug, Electrical - Pin Contact, No. 12, 14, & 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical: Socket Contact, No. 14 & 16 A.W.G., waterproof.
MS27145	- Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way No. 12, 14 and 16 A.W.G. Wateproof.
MS27148	- Contact; Pin, Electrical Connector: No. 12, 14, & 16 A.W.G.

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MS33800	Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35387	- Reflector, Indicating, Clearance.
MS35390	- Tire, Pneumatic: Low Platform Trailer, Earthmover, Grader, and Amphibious Vehicle.
MS35392	- Inner Tube: Pneumatic Tire.
MS35422	- Lamp Holder, Light, Marker, Clearance.
MS35423	- Light, Marker, Clearance - Service.
MS35424	- Light, Marker, Clearance - Blackout, 24 Volt.
Ms35643	- Gasket: Synthetic Ribber, Cap, Fuel Tank.
MS35644	- Filler Neck Assembly, Fuel Tank - Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35746	- Coupling, Automotive: Air Brake Lines.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS35865	- Axle, Vehicular, Ron-Driving, Single, for Tactical Wheeled Vehicles.
MS35866	- Axle Assembly, Vehicular Non-Driving, Tandem for Tactical Wheeled Vehicles.
MS39020	- Band, Markers Blank.
MS39062	- Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS39153	, Coupling, Quick Disconnect, 3- and 4-Inch, for Construction Equipment.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or steel.
MS51330	- Stop Light - Taillight, Vehicular - 24 volt, Blackout Tail, Blackout Stop.
MS51331	- Guard, Splash, Wheel.
MS51337	- Coupler, Drawbar, Ring - Heavey Duty.
MS51389	- Raincap, for Air Cleaner Intake, for Industrial Engines.
MS53004	- Valve, Relay, Emergency-Breakaway, Air Brake.
MS53007	- Plate, Identification - Emergency and Service.
MS53036 .	- Kingpin, Fifth wheel, 2 Inch.
MS53063	- Indicator, Air Cleaner, Intake: Restriction Mechanical Type.
MS75020	Connector, Plug, Electrical - 12 Contact, Inter-vehicular, 24 Volt, Waterproof'.
MS75021	Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

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DRAWINGS

ME

DL13216E9731 - CentraL Asphalt Plant.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form apart of this specification to the extent specified herein. Unless otherwise indicated, the issueMwe in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A48 - Gray Iron Castings.
A440 - High-Strength Structural Steel.
D5 - Method of Test for Penetration of Bituminous Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

RUBBER MANUFACTURES ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, Hew York, N.Y. 10022.)

AMERICAN TRUCKING ASSOCIATIONS, INC. (ATA)

National Motor Freight Classification Rules.
E-3 - Electrical Lighting and Wiring for Commercial Vehicle Bodies.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

DEPARTMENT OF TRANSPORTATION (DOT)

ICC Motor Carrier Safety Regualtions.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

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NATIONAL SAND AND GRAVEL ASSOCATION (NSGA)

Circular No. 80.

(Application for copies should be addressed to the National Sand and Gravel Association, 527 Munsey Building Washington, D. C. 2004.)

TIRE AND RIM ASSOCIATION, INCORPORTED (T&RA)

Tire and Rim Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., Comand Building, 34 North Hawkins Avenue, Akron, Ohio &333.)

ASSOCIATION OF ASPHLT PAVING TECHNOLOGISTS (ASPT)

Proceedings of the Association, Vol. 28.

(Application for copies should be addressed to the Association of Asphalt Paving Technologists, 1224 East Engineering Building, Ann Arbor, Michigan 48106.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedures.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N.Y. 10017.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC. (FMS)

Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, N. Y. 10017.)

VEHICLE EQUIPMENT SAFETY COMMISION (VESC)

Regulation V 3.

(Application for copies should be addressed to the Vehicle Equipment Safety Commission, 1026 17th street NW, Washington, D. C. 20036.)

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TRUCK-TRAILER MANUFACTUERS ASSOCIATION, INC. (TTMA)

RP #9 - Mounting of Required ICC Lighting Devices.
RP #14 - Heavy Duty Lighting System for Truck Trailers.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 1413 K Street NW, Washington, D. C. 20005.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.
Handbook H44 - Specifications, Tolerances, and Regulations for Commercial Weighing and Measuring Devices.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.9

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, chicago, Illinois 60606.) "

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N.Y. 10001.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)

B6.6 - Gear Tolerances and Inspection.
B55.1 - Specification for Multiple V-Belt Drives.

(Application for copies should be addressed to the United States of America standards Institute , 10 East 40th Street, New York, N.Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boilar and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers , 345 East 47th Street, New York, N.Y. 10017.)

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3. REQUIREMENTS

3.1 Description., The portable asphalt-mixing plant shall consist of all units, diesel engines, and other components, except heater fuel-storage tank, hot-oil heaters, asphalt melters, asphalt-storage tanks, and asphalt-storage transfer pump, required to result in a complete plant for continuously producing high-type asphaltic concrete mixes at a capacity of 100 to 150 tons per hour. The plant shall include but not be limited to the following units:

(a) Cold-aggregate feeder of hopper and frame construction used for blending and feeding predetermined amounts of aggregate to the cold-aggregate elevator.
(b) Cold-aggregate elevator of the centrifugal-discharge bucket type to receive cold aggregate from the feeder and discharge into the dryer.
(c) Aggregate dryer of single-rotating-drum construction to heat and dry the aggregate received from the cold-aggregate elevator.
(d) Fines (dust) collector of the exhaust fan and centrifugal cyclone-type separator to provide draft, remove products of combustion, and collect the fines emerging from the dryer.
(e) Two hot-aggregate elevators of the totally-enclosed, centrifugal-discharge, bucket type to receive the hot, dried aggregate.
(f) Gradation control unit of the 4-bin screening type to receive aggregate from the hot-aggregate elevator, separate and recombine the aggregate in the required proportions, and discharge into the hot-aggregate elevator serving the mixer.
(g) Fines feeder of the hopper-and-vane or screw-conveyor type to meter granular mineral filler into the hot-aggregate elevator feeding the pugmill.
(h) Portable conveyor of the elevating-troughing type for general use or to receive hot aggregate from the dryer and discharge into the mixer or single aggregate hopper when a hot-aggregate elevator is not wed for that purpose.
(i) Mixer of the continuous proportioning and mixing type to receive aggregate, spray, and mix with water or bitumen as required for various mixes.
(j) Piping system required for contral-plant erection and operation with tank car, melter, and transport unloading.
(k) Diesel engines.

The above units shall be mounted on semitrailers conforming to the weights and dimensions specified in 3.1.5 and maybe grouped at the discreteion of the supplier as long as the plant can be interconnected mechanically and operated at rated capacity under each of the following production arrangements:

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(1) Multiple aggregate with hot-aggregate elevators and gradation control,
(2 Single aggregate with hot-aggregate elevator or elevators.
(3) Single aggregate with portable conveyor.

The following exceptions relate to transportation:

(4 The cold-aggregate elevator maybe transported by truck or trailer.
(5) The hot-aggregate elevator from the dryer shall be cart-type trailer mounted. The hot-aggregate elevator from the gradation control unit shall be cart-type trailer mounted or may be mounted permanently on a trailer with other components.
(6) The fines feeder may be transported by truck or trailer or maybe mounted permanently on a trailer with other components.
(7) The portable conveyor shall be as specified in 3.21.
(8) Piping equipment and wood blocking shall be transported by cargo truck or trailer.
(9) The hopper and apron feeder (components of the mixer) maybe combined into a cart-type trailer at the discretion of the supplier.
(10) Burner, combustion chamber, and ignition cone maybe mounted on cart-type trailer.
(11) Dust collector engine maybe mounted on cart-type trailer.

3.1.1 Capacity. The output capacity of the mixing plant, complete with gradation control unit, shall be not less than that shown in table I.

Table I. Output Capacity

Aggregate	Percent passing square mesh screen		Percent initial surface moisture content (min)	Capacity (tons per hour) of high-type asphaltic concrete
	1-1/2 in.	1/4 in.		
Limestone	100	60	5	150
Limestone	100	60	9	100
		OR		
Gravel	100	70	5	150
Gravel	100	70	9	100

3.1.2 Stress limitations. The plant shall be constructed so that the nominal operation and transport static stresses in the load-bearing and load-carrying members shall not exceed 60 percent of the yield strength or 30 percent of the ultimate strength, whichever is smaller value. Under dynamic travel conditions with unimproved roads imposing twice the maximum vertical

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acceleration force (2 g's), the sum of the static stresses plus the dynamic stresses shall not exceed 100 percent of the yield strength or 50 percent of the ultimate strength, whichever is the smaller value. This anatysis shall include calculations and diagrams of the following for each unit furnished to meet the requirements specified in 3.1:

(a The load distribtion diagram.
(b The vertical shear diagram.
(c! The bending moment diagram.

3.1*3 Compatibility. Without modification, each component of the mixing plant shall assemble and operate in conjunction with other components to form 8 central in all the arrangements as specified in 3.1.

3.1.4 **Piping-system, wood-blocking, and erection drawings.** **The supplier shall provide a complete set of piping-system, wood-blocking, and erection drawings for all central plant arrangements as specified in 3.1 and 3.23. The drawings prepared by the supplier shall be similar in arrangement, content, and scope to DL13216E9731. Original drawings shall be pencil on high-quality reproducible paper. Originals and three sets of prints shall be** supplied.

3.1.5 Weights and dimensions. The gross vehicle weight (m) of each unit shall include the weight of the trailer or semitrailer with all machinery, equipment, accessories, attachments, and lubricants. Component and vehicular ratings shall not be raised to conform to the requirements of this specification. The physical characteristics of each unit prepared for transport shall be as specified in table U, with the unit on level ground and with the inflation pressure of the tires as recommended in the TRA Yearbook for the imposed GVW loading.

Table II. Weights and Dimensions

Characteristics	Requirements
Overall length	**45 feet, max.**
Overall width	**120 inches, max.**
Overall height	**150 inches, max.**
Ground clearance, exclusive of suspension	**12 inches, min.**
Semitrailer:	
GVW	55,000 lb, max.
Fifth-wheel load	15,000 lb, max.
Axle load, each	18,000 lb, max.
Upper fifth-wheel height range	54 to 56 inches
Kingpin location, range (SAE J701)	24 to 36 inches
Swing radius clearance (SAE J701)	57 inches, max.
Landing wheel Clearance (SAE J701)	84 inches, min.

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Table 11. Weights and Dimensions (cont'd)

Characteristics	Requirements
Cart-type trailer:	
GVW	14,000 lb, max.
Vertical lunette load, range	10% to 12% of GVW
Axle load, each	10,000 lb, max.
Tow bar eye height, range (SAE J849)	31 to 37 inches, max.

3.2 Preproduction model. The supplier shall furnish a mixing plant for examination and test within the time frame specified to prove, prior to starting production; that his production methods and choice of design detail will produce mixing plants that comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Initial production. When specified, the supplier shall furnish to the Government one or more mixing plants for inspection as speciflid in 4.4.

3.4 Material. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. All materials shall be new and unused.

3.5 Safety. The safety characteristics of the plant shall provide necessary safety and protective devices or prominent warnings without impairing the operating functions of the plant. For exposed components and systems which are subject to high temperatures, high pressures, electrical actuation, or which are inherently hazardous, the plant's safety characteristics shall provide correct safeguarding enclosing, and insulating features. All step and platform surfaces shall provide a nonskid footing. Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be eliminated where injury could be caused to operating or maintenance personnel. Humeri factors engineering characteristics shall be incorporated into the plant. The operating characteristics of the plant shall provide for operation by personnel ranging from the small man clothed through the large man arctic clothed, in accordance with SAE J833.

3.6 Maintainability. The plant shall opera" as specified herein without maintenance other than the supplier's recommended schedule of lubrications and adjustments as established by a lubrication chart, and a maintenance schedule submitted by the supplier prior to the test. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed

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attachments. Covers or plates, such as those on shrouds or housings, which must be removed for adjustment or maintenance shall be equipped with quick-disconnect fastenings. Disconnect plugs and receptacles, multiple-line connectors in the electrical system, and detachable- and attachable-type fittings in hydraulic and pneumatic system, as applicable, shall be utilized where practicable to permit component removal and replacement. All disconnect points shall be clearly indicated. Maximum practicable use shall be made of interchangeable hardware and fastening devices with a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common (standard) parts. Lockwashers shall be secured to bolts or screws except where this practice is not practicable. Drain outlets shall be located for accessibility and shall permit the use of receptacles for collecting drainage. Drain plugs shall be removable by wrench. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

'3.7 Transportability.

3.7.1 Performance characteristics. Each vehicle shall meet the transport performance characteristics specified herein when: (1) prepared for transport; (2) completely serviced and adjusted as specified in 3.8; and (3) when towed by a 5-ton, 6 by 6, M52 truck-tractor for semitrailers or any 5-ton, 6 by 6, M39 series truck for cart-type trailers.

3.7.1.1 Highway operation. Each vehicle shall operate over smooth, hard-surfaced primary highways at speeds up to 20 mph.

3.7. Unimproved road operation. Each vehicle shall operate over unimproved roads at speeds up to 10 mph.

3.7.1.3 Trailing and turning ability. Each vehicle shall trail the towing vehicle without weaving. Each vehicle shall fallow the towing vehicle without exceeding 75 percent of the ICC towed-vehicle tolerance. Jackknife turns up to an angle of 90) degrees for semitrailers and turns up, to 60 degrees for cart-trailers and dolly semitrailer combinations, as measured from straight-ahead to the right and to the left, shall not cause damage to or intesference between the towed vehicle dolly and the towing vehicle.

3.7.1.4 Braking ability. On each vehicle the combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in accordance with ICC requirements, except that the stopping distance shall be accurately measured from initial movement of the brake pedal to stopping point. When actuated, the emergency breakaway system shall produce smooth emergency brake application. On multiple-axle units, wheel hop during braking shall be prevented.

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3.7.1.5 Parking ability. When the emergency parking brakes are applied, the wheels shall remain locked on grades up to 40 percent despite the depletion of the compressed air supply.

3.7.2 Rail transports. Each vehicle shall withstand impact forces encountered in rail transporr witout breakage or permanent deformation. Sectionalization for rail transportiation shall be within the dimensional limits shown on figure 1. Sectionalization and reassembly of each unit listed in 3.2 shall each be accomplished in not more than 6 hours without cutting or welding.

3.7.3 Air transport. Sectionalization for air transportation shall be within the dimensional limits shown on figure 2. Sectionalization and reassembly shall be accomplished as specified in 3.7.2.

3.8 Vehicle servicing and adjusting. Prior to delivery of the vehicles, the supplier shall adjust and service each vehicle for travel use including at least the following: Adjust brake system; check electrical system; inflate all tires; and completely lubricate the chassis and components as specified herein.

3.9 Vehicular components. The cart-trailer and semitrailer components shall be as specified herein.

3.9.1 Intervehicular connections.

3.9.1.1 Truck-tractor and semitrailer. The semitrailer shall be furnished with airbrake couplings as specified in 3.9.7 'and the electrical intervehicaler receptacle as specified in 3.9.9 for connecting the truck-tractor airbrake and electrical systems to the semitrailer systems. The couplings and the receptacle shall be installed on the front of the semitrailer, symmetrical about the centerline in accordance with SAE J702. Clearance about the couplings and the receptacle shall permit connection of the truck-tractor hoses and harness to the semitrailer without interference with the upper fifth-wheel plate or other components. The airbrake couplings shall be protected with metal spring dirt covers. The electrical receptacle shall have pigtail leads terminating within junction boxes or a splashproof, semitrailer-type, nosebox. The nosebox shall also house the three circuit breakers as specified in 3.9.9 for the protection of the service-lighting circuits.

3.9.1.2 Truck and cart-trailer. Each trailer shall be furnished with an intervehicular electrical harness as specified in 3.9.9. with airbrake lines as specified in 3.9.7.2 for connecting the trailer to the truck, and with two safety chains. All intervehicular components shall be installed on the trailer so that they shall extend between 36 and 42 inches toward the prime mover from the center of the lunette eye. A plug clip shall be furnished

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to hold the electrical harness plug and to double the harness back on the trailer tongue. Two dummy couplings shall be furnished and fastened to the tongue for doubling back and holding the lines. The intervehiculer line routing, connected and stowed, shall be such that dropping of the tailgate of the military truck (see 3.7.1) will not damage any lines. A semitrailer-type, electrical nosebox shall be furnished (see 3.9.1.1).

3.9.2 Chassis frame. The framing members of the vehicle shall support the weights and maintain alinement under all conditions of transport and plant operation as specified herein, and shall provide a rigid base for the operation of machinery. All parts of the chassis frame where welding is required shall be electrically welded.

3.9.3 Semitrailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a 36-inch-diameter lower fifth wheel and shall extend the entire width of the gooseneck. The plate shall be fabricated of high tensile steel with not less than 50,000 psi yield point, and shall have a thickness of not less than 0.370 inch. Both longitudinal and transverse reinforcing members shall be provided. At the front of the plate, a front coupling ramp shall be furnished. Ramps that are not flush with the front crossmember shall include at least four vertical gussets and shall be provided with holes for water drainage. An SAE 2-inch-throat-diameter kingpin conforming to MS53036 shall be installed not less than 24 inches or more than 36 inches from the front of the semitrailer. The center of the kingpin shall be within 1/8 inch from the longitudinal centerline of the semitrailer.

3.9.4 Cart-trailer towing attachment. Each cart-type trailer shall be equipped with stowing attachment having a lunette conforming to MS51337 installed in accordance with SAE J849. The lunette shall be supported the entire length of the shank in a housing integral with the towing attachment. The clearance between the shank and the housing shall not exceed 0.035 inch. Upper and lower stops shall hold the square flange of the lunette to prevent rotation. The lunette shall be installed on the towing attachment within 1/8 inch on either side of the centerline of the trailer. The towing attachment may be either fixed or removable type, constructed and installed to provide a structure necessary to resist pulling, bending, and twisting without permanent deformation a damage under all transport conditions specified herein. The towing attachment shall be equipped with tire rub plates on both sides. These plates shall be curved top and bottom to prevent possible interference with the towing Vehicle's tire lugs. Two safety chains conforming to ICC regulations shall be furnished. The safety chain hooks shall pass through the shackles of the military truck specified in 3.7.1.

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3.9.5 Landing jacks and jacklegs. Landing jacks shall be provided on semitrailers and jacklegs on cart-type trailers for raising, leveling, and supporting each vehicle, except hot-aggregate elevators and portable conveyor. The cart-type trailer for the hot-aggregate elevator shall be provided with jacklegs for support only. The landing jacks and jacklegs shall be manually operated screw jacks or jacks with screw type equalizing feet, and shall be permanently installed on the vehicle. The landing jacks and jacklegs shall be of such strength, design, and location that coupling and uncoupling the trailer with a towing vehicle will not cause stresses to exceed the limit specified in 3.l.2. Operating jacks shall be furnished to raise each vehicle into position for plant operation. The operating jacks shall be either pemanently attached or removable* When the jacks are removable, provisions shall be made for stowing the jacks on the vehicle. operating jacks shall be of' such strength, design, and location that operation or the mixing plant under load till not cause stresses to exceed the limit specified in 3.1.2. All jacks shall be equipped with square, metal, mud pads having a ground bearing area of not less than 144 square inches, and shall be interchangeable between all vehicles. Any jack may serve as a combination landing jack or jackleg and operating jack, providing it meets all requirements specified herein.

3.9.6 Running gear. The vehicle running gear shall be the stringless type on cart-trailers and semitrailers. The axle, or the forward axle on tandom axles, shall be in alinement with the longitudinal centerline of the vehicle so that the two diagonal measurements from the center of the lunette or kingpin to the outer center of each hub do not differ in excess of 1/16 inch. The rear axle of tandem-axle asseblies shall be in alinement with the front tandem axle so that the parallel measurements between hub centers on each side do not differ in excess of 1/8 inch. Stub axles or through axles may be furnished. The length of axle tubes or through axles may be increased during manufacture providing the mounting pads are increased at least an equal distance to insure that the structural rigidity is equal to that of the axles shown on the standards and drawings. Each cart-type trailer and semitrailer shall be equipped with wheel splash fenders and splash guards which shall restrict water and other road-surface substances thrown from the wheels. These components shall be in accordance with SAE 3682, SAE J683, and MS51331.

3.9.6.1 Tandem-axle suspensions. Suspensions shall withstand the imposed loads without overload, permanent defamation, war, failure, or malfunction. The components and the installation shall be as recommended by the suspension manufacturer for use in severe, abnormal service. The mounting seats shall be installed on the axle in accordance with the recommendations of the axle manufacturer. Tandem-axle assemblies shall have divergent angular movement

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between the front axle and the rear axle. This movement shall permit not less than 8 inches difference in elevation between adjacent wheels on each side as measured from the level vehicular structure l Rubber bumpers shall be provided to cushion the axle at full jounce. For tandems, the suspension configuration shall nullify brake-hop during all degrees of brake application.

3.9.6.2 Axle assemblies. The axle assemblies and components of these assemblies are the permissible options to be incorporated in the vehicles of the mixing plant. At the discretion of the supplier, similar standard commercial axles with equal or greater axle manufacturer's load rating are acceptable. The selection, quantity of axles, and location shall be at the discretion of the supplier, and the load on any one axle shall not exceed the limit specified in table II for each type trailer.

3.9.6.2.1 MS35865-A axle assembly. The axle shall conform to MS35865, dash number Al, A2, A4, or A5. The maximum axle load shall be limited to the sum of the tire-capacity ratings l The axle shall be compatible with the air-over-hydraulic brake system as specified in 3. 9.7.1. Except as permtted in 3.9.6.2.3, the axle shall be equipped with single wheels, tires, and inner tubes. Tires and wheels shall conform to the standards listed on MS35865 for the axle furnished. Inner tubes shall be selected from MS35392 to fit the tires. The foundation brakes for each axle shall not not less than . 330 square inches of' swept braking surface area.

3.9.6.2.2 MS35865-C axle assembly. The axle shall conform to MS35865, type C, dash number optional, except that the straight-air-brake system shall be furnished in lieu of the air-over-hydraulic brake system The maximum axle load shall be limited to 18,000 pounds, Except as permitted in 3.9.6.2.3, the axle shall be equipped with duel wheels, tires, and inner tubes. Tires and wheels shall conform to the standards listed on MS35865, type C, Inner tubes shall conform to MS35392-15. The foundation brakes for each axle shall have not less than 560 square inches of swept braking surface area. The foundation brake assemblies shall be self-adjusting, right angle actuated type.

3.9.6.2.3 Wide-base tires and wheels. Single wide-base tires and wheels may be substituted for the tires, inner tubes, and wheels sqecified in 3.9.6.2.1 and 3.9.6.2.2. All components shall be in accordance with the TRA Yearbook end Supplementary Service Data for 30 mph speeds. MS35865, type A axles shall be equipped with wide-base tubeless tires, size 12 by 16.5, 10 ply rating, having traction lug tread design. Disc wheels shall be size 9.75 by 16.5, 3/4-inch positive offset, of the drop-center 6-hole type conforming to SAE J694 , having an 8-3/4-inch bolt circle . MS35865, type C axles shall be equipped with wide-base tubeless tires, size 16.5 by 22.5, 16 ply rating, having traction lug tread design. Disc wheels shall be size 13.00 by 22.5, industry standard positive or negative offset, of the 10-hole type

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conforming to SAE J694, having aN 11-1/4-inch bolt circle. Valves and valve caps shall be of domestic manufacture and shall be installed in accordance with the TRA Yearbook. Valve caps shall be of metal and shall be equipped with elastomer inner seals.

3.9.6.2.4 Low-platform-trailer axle. The axle components shall be interchangeable with the major assemblies shown an MS35866-DL. The maximum axle load shall be limited to 18,000 pounds with dual tires. The brake system shall be of the straight-air type as specified in 3.9.7.2. Foundation brakes shall be not less than size 12-1/4 by 7-1/2, Commercial wheel assemblies and military tires shall conform to the following:

(a) Wheel assemblies shall be malleable iron or cast steel. Wheels shall be the 15-inch, 6-spoke type with bearing cups pressed in place. Malleable wheels shall be SAE J432, grade 35018. Cast steel wheels shall be SAE J435, grade 0030. Rim clamps shall be SAE J432, grade 35018. Studs and bolts shall be in accordance with SAE J429, grade 5, and shall have a minimum tensile strength of 120,000 psi. Brake drums shall be of cast iron SAE J431, SAE No. G 4000 or G 3500. The demountable rims shall be 7.5 by 15 size, 5-degree, 3-piece for tube-type tires.

(b) Tires shall be 11.00 by 15, 16-ply rating, conforming to MS535390-47.

(c) inner tubes shall be 11.00 by 15 conforming to MS35392-26.

3.9.6.3 Tires. Tires shall not be loaded in excess of the values specified. The tires shall be free of majar defects when inspected in accordance with MIL-STD-1224.

3.9.7 Brake system. The semitrailer shall be equipped with a waterproof, internal expanding, straight airbrake system in accordance with ICC Motor Carrier Safety Relations, sections 193.40 through 193.43 and 193.45 through 193.50, except as otherwise specified herein. The system shall include service and emergency airbrake line couplings, service and emergency line filters, relay emergency valve, air reservoir, cylinders or chambers, foundation brake assemblies on each wheel, and all other plumbing and components required for a complete, matched airbrake system Airbrake line coulings shall conform to MS35746 and to SAE J702, service on the curb side and emergency on the road sides and shall be of the polarized or noninterchangeable types. Each coupling shall be equipped with a spring-loaded dust cover. The couplings shall be color coded blue for service and red for emergency. In additions line identiftcation tags conforming to MS53007 shall be furnished. The relay emergency valve shall conform to MS53004, except that the wrapid bleed-down valve shall be replaced with a plug on the straight airbrakes. Conformance of the relay valve to ICC regulations is mandatory. Each air reservior shall

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conform to SAE JIO. The service brake system shall have a total air storage capacity of not less than two times the recommended air volume for truck-trailers in SAE J813. The reservoir shall be equipped with a remote control or a 90-degree twist drain valve mounted on the low end of the reservoir and positioned on the curb side of' the semitrailer accessible to personnel from the standing or kneeling 'position. The size of the foundation brakes on each axle shall be the same on each vehicle and the swept drum total braking surface of each axle (two spindles) shall be as specified for the axle selected. The foundation brake components shall not be overstressed or exceed the brake component manufacturer's ratings when a full service brake application is performed. All components of the brake system shall be located, installed, and protected to prevent damage or malfunction from water, ground obstructions, and debris and to obtain maximum efficiency and ease of maintenance. No part of' the braking system shall extend below the bottom of the rim of the wheels. An air-over-hydraulic brake system shall be furnished when an axle assembly as specified in 3.9.6.2.1 is used. A straight-air-brake system shall be furnished when an axle assembly as specified in 3.9.62.2 or 3.9.6.2.4 is used.

3.9.7.1 Air-over-hydraulic brake system. The air reservoir shall have a volume capacity of not less than 1,750 cubic ijnches. A power cluster shall 'be required for each axle on the cart-type trailer. Multiple-axle trailers shall be provided with either axle-by-axle or brake-by-brake protection. The power clusters shall be matched to the size of the foundation brakes they operate and shall have their high mortality parts interchangeable with other military-type trailers or semitrailers.

3.9.7.2 Brake lines. The brake lines of the plant shall be routed within the protection of the framing members and shall be secured by cushioned support clamps at intervals not greater than 3 feet. At each point when the brake lines pass through metal member, the lines shall be protected by a rubber or nylon grommet and secured, or a through-frame connector shall be used. The routing and connection locations shall facilitate replacement of lines from bulk supply. Not less than 5 inches of slack in flexible air hose shall be provided at each of the relay emergency valve connections to permit interchange of the various makes of MS53004 valves. The ends of these lines shall be identified by markers conforming to MS39020 installed near the fittings. The brake line components shall conform to the following:

(a) Flexible air line - MIL-H-3992, type I, or SAE J4O, SAE 40R2, type E or F, having reusable fittings.
(b) Rigid air line - SAE J844 copper tubing.
(c) Fittings - SAE J512, heavy-duty, airbrake compression type, reusable.

At the suppliers option, SAE J40, SAE 40R2, type E or F, air hose may be furnished in Lieu of (a) and (b) above.

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3.9.7.3 Brake lining. The brake lining shall be of the metallic type having nonfading characteristics when subjected to heat and to water immersion. The lining shall be coded class FE in accordance with Regulation V-3 of the Vehicle Equipment Safety Commission. The lining selected by the supplier shall be of a stanard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc. The lining segments shall be not less than 1/2 inch nominal thickness. The brake lining shall be installed on the shoes by the use of hardened steel bolts and nuts as recommended by the lining manufacturer. The brake drums shall have a wearing surface of automotive gray cast iron and shall be of the anti-bell mouth type.

3.9.7.4 Brake preservation. All metal surfaces enclosed by the brake drum, including the interior surface of the drum, metal surfaces of the shoes, surfaces of backing plates and all attached parts, shall be coated with primer conforming to MIL-P-46093 or TT-P-664. Ramps, wedges, pins, eccentrics, cam levers, anchors, and retracting springs shall be coated with preservative oil conforming to MIL-L-21260, grade 2. Items or surfaces which are wholly prime coated, painted in accordance with Government specifications, or constructed of materials providing corrosion protection equivalent to that provided by MIL-C-16173, grade 1, do not require additional protection.

3.9.8 Chock blocks and cradles. Two wood chock blocks and two wire cradles shall be furnished for each axle assembly. The cradles shall be located near the wheels, on the front of the fenders, when practicable. The chock blocks shall be captive chained to the cradles on the outer side with chains of sufficient length to permit placement of the chock blocks either in front of or behind the vehicle's wheels. The cheek blocks shall be as shown in figure 3 and the cradles shall be as shown in figure 4.

3.9.9 Vehicular electrical system. Each semitrailer and cart-type traile shall be equipped with a waterproof, 24-volt vehicular electrical system for service and blackout "night travel. Except as specified herein, the electrical system shall be in accordance with the following requirments:

(a) SAE Handbook, Electrical Equipment and Lighting Section.
(b Federal Motor Vehicle Safety Standards.
(ICC Motor Carrier Safety Regulations.
(d) TTMA Recommended Practice RP #9 and RP #14.
(e) ATA E-3.
(f) Figure 5.

All components and wiring shall be so located and guarded as to damage and permanent deformation, and provide maximum protection from shock and vibration. The components and wiring shall also be accessible for replacement without disturbing permanently installed housings and guards. The electrical system shall be in accordance with tablee III. Stem quantities that are not specified shall be furnished as required (AR) for each unit.

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Table III. Vehicular Electrical System

Description	Requirement	Quantity
Service Lights: Clearance Side-Markers and Identification	MS35423	14 ea.min.
Blackout Lights: Clearance and Side Markers	MS35424	8 ea.
Installation	Horizontal, using information on MS35422	AR
Service Combination Lights	Commercial SAE STI, 4 in. dia. waterproof, flush mount type (SAE J759)	2 ea.
Blackout Combination Lights	Military MS51330	2 ea.
Reflectors	MS35387	8 ea.min.
Front Semitrailer Receptacle Assembly	MS75021-1 with matching 4-hole spring loaded cover	1 assembly
Nosebox or Junction Boxes	Commercial, Watertight types, see figure 5	AR
Intervehicular Harness for Cart-Trailer Type Units only	MIL-C-13486/12, length as specified, equipped with plugs conforming to MS75020 on each end, mating towing vehicle to unit	1 assembly
Low-Tension Cable	MIL-C-13486, 14 AWG min	AR
Harness Insulation	MIL-I-631, MIL-I-15126	AR
Loom	Semitrailer type polyethylene tubing or oilproof SAE J562 loom	AR
Conduit	Commercial, thinwall type	AR
Cushioned Support Clamps	MS21322, MS21333 & MS21334	AR
Circuit Number Bands	MS39020	AR
Friction Connectors	MS27142 thru MS27148 and MS33800 (waterproof)	AR
Connector Lubricant	MIL-S-8660	AR
Connector Holder Clip	Plastic or metal spring clip conforming to or replaceable by MS17160-29	AR
Terminal Connector	Spade type SAE J561 with heat-shrinkable insulator sleeves	AR
Circuit Breakers	MS39062	3 ea.

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3.9.9.1 Lighting. All lights shall operate as specified herein on 24 volts. All lights shall be provided with shock and vibration protection by location and though use of resilient mountings to prolong filament life. Front and rear identification, clearance, and side-marker lights shall be cluster mounted on plates inserted into lamp bracket sockets conforming to SAE J725. These clusters shall be removable for storage during periods of operation. Each cluster shall be equipped with instruction plates to preclude incorrect replacement of the cluster in preparation for travel. The rear service combination lights shall be mounted in industry standard 4-1/2 inch diameter holes. These lights shall be of the flush or recess mount type having either a sealed unit mounted in an elastomer ring, 02 a bulb and lens mounted in an elastomer watertight housing. If furnished, bulbs shall be double contact, indexing base type, Trade No. 1662. These lights shall be installed as far apart as possible for turn-signal operation as well as stop and taillight indication. The blackout combination lights shall be installed inboard or below the service combination lights. These lights shall be installed and centered behind 6-inch-diameter holes and shall be flush with the service combination lights. Forward shielding shall be provied to prevent damage from road splash.

3.9.9.2 Wiring. The wire for both service and blackout light systems shall be as shown on figure 5. The wiring shall be protected from heat and road splash, and routing shall be within framing members wherever possible, The electrical connector receptacle shall be located on the front as specified in SAE J702. The leads from the receptacle shall routed to a splash-proof, semitrailer nosebox or to the twin junction boxes. The nosebox shall contain terminals for all circuits and the three circuit breakers for the designated circuits. The terminals shall be either spring loaded for bare wire cables or nut and washer for spade terminal cables. Cushioned clamps, loom, thinwall conduit, and grommets shall be provided to prevent whipping, chafing, and other mechanical and heat damage. Wiring shall be fastened to the framing members at intervals not to exceed 3 feet. Two or more cables in the same routing shall be enclosed in loom having each end waterproofed, or shall be bound together in harness with one-half overlapping wraps of plastic electrical tape. Splices shall be waterproof'. Mechanical splicing devices, it used shall be fastened to the framing members with the interior of the device waterproofed with silicon lubricant. All circuits shall be identifed by number coding as specified in figure 5. The circuit number bands shall be installed near each connection. The mating surfaces of the elastomer friction connectors shall be lubricated with silicon lubricant prior to assembly. The assembled friction connectors shall be secured to the framing with holder clips to prevent fatigue cable breakage. Terminal connectors shall be of the straight-type spade or side-type spade, and application shall be limited to use within the waterproof nosebox and as component ground leads.

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3.9.9.3 Reflectors. Reflectors shall be mounted in accordance with ICC requirements. In addition to ICC requirements, two amber reflectors shall be mounted on the front of the vehicle at approximately the same height as those mounted on the sides.

3.10 Bearings, Unless otherwise specified herein, all bearings shall be greaseable antifriction type conforming to FF-B-171, FF-B-185, or FF-B-187, as applicable, Mountings of bearings shall permit split tapered sleeve and locking nut or inner race secured to the shaft by a minimum of two set screws.

3.11 Liners. Liner plates shall be installed in areas of all hoppers, bins, chutes, spounts, and other metal surface areas subject to continual abrasive wear caused by aggregate movement. Liners shall be replaceable high-carbon steel plates with not less than 0.35 percent carbon content. All bolt heads subjected to the flow of material shall not extend beyond the surfaces exposed to the material and shall be of hardened steel.

3.12 Drives, All the necessary drive gears, gearboxes, shafts, univwersal joints, clutches, bearings, and other components required to operate the drive system as specified herein shall be provided. Unless otherwise specified herein, all drives shall be of the mechanical type.

3.12.1 Chain drives. Chains used in drive trains shall be in accordance with MIL-STD-421, MIL-STD-423, or MIL-STD-424, and shall have provisions for adjustment.

3.12.2 V-belt drives. V-belt drives shall be in accordance with USA Standard B55.1 and shall have provisions for adjustment.

3.12.3 Gears and gearboxes. All gears shall be enclosed in gearboxes and shall operate in an oil bath. Gears shall conform to USA Standards B6.6, **class 2 or 3. Gearboxes shall be provided with oil seals to prevent loss of oil and entry of foreign material.**

3.12.4 Line shafting. All line shafting shall be of carbon steel containing not less than 0.40 percent carbon.

3.12.5 JAW clutches. The entire mixing plant shall be provided with jaw-type clutches to engage and disengage components as necessary for operation of the mixing plant in the various arrangemennt as specified in 3.1 without shutting down an engine. Control levers shall be provided with positive-position locking devices. A schematic diagram of' the mixing plant indicating location of clutches shall be provided and shall be subject to the approval of the contracting officer.

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3.12.6 Overload safety devices. Replaceable shear collars or the equivalent shall be provided in the drive lines or component throughout the entire mixing plant in all locations where overload may cause damage.

3.12.7 Universal joints. All universal joints shall be of the antifriction bearing type.

3.13 Engines. The engins, including all systems, components, accessories, and auxiliaries shall conform to MIL-E-11276, except as otherwise specified herein. The number of engines furnished per plant shall be at the discretion of the supplier, except not more than four may be provided.

3.13.1 Power and speed rating. The power and speed rating of the engines shall be such that operation of the plant under any of the operational requiremarts and conditions specified herein, will not require horsepower in excess of the continuous horsepower rating of the engines for the applicable governed speed, as established by the Qualified Products List, QPL-11276.

3.13.2 Fuel tank. Each engine shall be furnished with a fuel tank conforming to MIL-T-46786, type II, class 2. The "fuel tank shall have a total minimum capacity sufficient for not less than 10 hours operation in any operating condition specified herein. Fuel capacity shall be based on 100 percent of the fuel consumption rate obtained during conduct of tests in test method series 2000 of MIL-STD-1400, at the applicable horsepower and governed speed.

3.13.3 Oil filters. The engine oil filters shall be of the full flow type in accordance with MIL-E-11276.

3.13.4 charging generators. The battery charging generators shall be 18 ampere capacity in accordance with MIL-E-11276.

3.13.5 storage batteries. Each engine shall be provided with storage batteries and a battery box in accordancee with MIL-E-11276.

3.13.6 Air cleaner. The air cleaners shall be of the dry type, conforming to MIL-A-52363, class II, and a rain cap conforming to MS51389 shall be provided. The air cleaners shall be provided with a restriction indicator conforming to MS53063-2 if located remotely or MS53603-3 if mounted on the air cleaner.

3.13.7 Engine housing. Engine housings in accordance with MIL_E-11276, shall be furnished. Engine housing side panels shall have not less than two handles or handholds to facilitate opening, removal, and replacement of panels.

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3.13.8 Instrumentation. Each engine shall be furnished with an instrument panel in accordance with MIL-E-11276. The panel shall include the lubricating oil pressure gages coolant temperature gage tachometer, ammeter, engine hour-meter, and fuel gage. The hourmeter and tachometer may be combined into one unit.

3.13.9 Clutch. Each engine shall be equipped with a clutch. The clutch shall be accessible for lubrication, adjustment and replacement of parts. The clutch lining shall conform to KKK-L-370, type II, and shall be installed in accordance with the lining manufacturer's instructions. Linings which meet or exceed the performance requirements of KKK-L-370, type II are also acceptable. The clutch design shall be such that engagement or disengagement at any engine speed up to maximum governed speed and under any operating condition specified herein, shall not cause damage to the clutch. The clutch capacity shall be not less than 150 percent of the maximum torque available from the engine as determined by test method series 2000 of MIL-STD-1400.

3.14 Cold-aggregate f'ceder. The cold-aggregate f'ceder shall be of hopper and frame construction and shall consist of a two-bin coarse-aggregate hopper, a two-bin fine-aggregate hopper, a reciprocating or belt-conveyor feeder (course-aggregate hopper), and a belt-conveyor feeder (fine-aggregate hopper) with discharge enclosure, all supported a a rigid steel frame end mounted on a pneumstic-tired semitrailer chassis. The cold-aggregate feeder shall discharge into the cold-aggregate elevator or portable conveyor l

3.14.1 Capacity. The cold-aggregate feeder shall have a discharge capacity of not less than 150 tons per hour (tph) when charged with limestone or gravel.

3.14.2 Hoppers. The coarse-aggregate and fine-aggregate hoppers shall be of welded-steel construction, each divided two two bins to permit handling of two sizes of aggregates in each hopper (total of four sizes of aggregates)* The bin side p1ates and. divider plates shall prevent bridging or clogging of the aggregate. Each bin shall have an opening of not less than 7 feet by 9 feet at the top. Hopper and bin extensions shall be provied for both hoppers. Bins and bin extensions shall be provided with angle iron reinforcing to prevent damage to the edges while bins are being charged. Each bin shall be provided with an individual, adjustable control gate to regulate the flow of aggregate. Each control gate shall be provided with an adjustment lever and postion looking device.

3.14.2.1 Coarse-aggregate hopper. The coarse-aggregate hopper shall be located at the rear of the semitrailer chassis and shall discharge onto a reciprocating or belt-conveyor feeder. The struck capacity of the coarse-aggregate hopper (without extensions) shall be not less than 18 tons (9 tons each bin). Hopper and bin extensions shall increase the capacity of each bin by not less than 6 tons.

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3.14.2.2 Fine-aggregate hopper. The fine-aggregate hopper shall be located at the front end of the semitrailer chassis and shall discharge onto the belt-conveyor feeder. The struck capacity of the fine-aggregate hopper without extensions shall be not less than 25 tons (12-1/2tons each bin). Hopper and bin extensions shall increase the capacity of each bin by not less than 6 tons.

3.14.3 Reciprocating feeder. When a reciprocating feeder is furnished instead of a belt-conveyor feeder for the coarse-aggregate hopper, the reciprocating feeder shall be of the pan type and shall be not less than 24 inches wide. The feeder shall permit the handling of two sizes of aggregates and shall receive material from the coarse-aggregate hoppers.

3.14.4 Belt-conveyor feeder(s). The belt-conveyor feeder shall consist of a suitable steel frame, conveyor belt, head-end (drive) pulley, tail-end pulley, carrier rolls, and return rolls. Straight carrier rolls shall be mounted under the belt to support the aggregate as it is discharged from the fine-aggregate hopper. Steel dust pans shall prevent dust and aggregate from falling onto the backside of the conveyor belt. Steel side plates and rubber flashing shall be provided to prevent spillage of aggregate from the sides as it is discharged from the hopper. A steel enclosure shall be provide at the discharge end-of the belt feeder to direct the aggregate to the cold-aggregate elevator. Pulleys and rollers shall be mounted on steel shafts and supported by antifriction bearings.

3.14.4.1 Conveyor belt. The conveyor belt shall conform to ZZ-B-206, type A, class 2, grade 375, 30 inches in width, with the following exceptions: The top cover shall be 1/8 inch thick and the bottom cover shall be 1/16 inch thick with a skin coat between all plies. The tensile strength of the top and bottom covers shall be not less 2,500 psi. The friction pull between plies and cover shall be not less than 16 pounds. The ends of the belts shall be laced with metal fasteners suitable to the type belt supplied and to the diameter of the pulleys used. Conveyor belt in accordance with RMA grade 2, 2-ply all nylon carcass, 30 inches in width is also acceptable.

3.14.4.2 Belt scraper. A belt scraper shall be provided to prevent buiZ& up of meterial on the feeder belt-conveyor return rolls. The -scraper shall be single-rubber-blade, spring-loaded, replaceable. The blade shall be of neoprene rubber not less than 1 inch thick with a tensile strength not less than 1,200 psi and a durometer hardness between 65 and 75.

3.24.4.3 Bearings. Head and tail-end pulley shafts shall be furnished with self-alining bearings and shall be be provided with labyrinth seals.

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3.14.4.4 Takeup adjustment. The belt-conveyor feeder(s) tail-end section shall be provided with a takeup adjustment mechanism to obtain belt tension. The takeup mechanism shall consist of two acme thread arrangements (one per side). The total takeup shall be not less than 12 inches, including not less than 3 inches negative takeup.

3.14.4.5 Belt carriers. The cold-feeder conveyor belt carrier rolls shall conform to MS16961 Return rolls shall conform to MS16964.

3.14.4.6 Head pulley. The cold-feeder head pulley shall be constructed of steel, crowned, provided with lagging, and shall be not less than 8 inches in diameter. The outside surface of the pulley prior to affixing lagging shall be concentric with the shaft within a total indicator reading of 0.060 inch, except a total indicator reading of 0.090 inch is permissible in the area immediately adjacent to a longitudinal weld.

3.14.4.6.1 Crown. The crown shall be approximately 1/8 inch per foot taper to the center of the pulley.

3.14.4.7 Tail pulley The cold-feeder tail pulley shall conform to the requirements for the head pulley except that lagging will not be required. The pulley shall 'be self-cleaning.

3.15 Cold-aggregate elevator. The cold-aggregate elevator shall be a partially enclosed, centrifugal-discharge, bucket-type, aggregate elevator with discharge spout, drives, and gearboxes. Means for ajusting and maintaining alinement and tension of the driving mechanisms shall be provided.

3.15.1 Capacity. The cold-aggregate elevator shll have a discharge capacity of not less than 180 tph when **supplied with limestone or gravel.**

3.15.2 Bucket line.

3.15.2.1 Buckets. The buckets shall be steel of a thickness not less than 1/4 inch. Each shall have a capacity of not less than 1/2 cubic foot. Buckets shall be installed by bolting on the bucket-line chain with not less than four steed bolts.

3.25.2.2 Bucket-line chain. The bucket-kine chain shall be of the roller-chain type. The maximum design load shall not exceed 15 percent of the ultimate tensile strength of the chain or chains. The pitch and number of chains shall be at the discretion of the supplier.

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3.15.2.3 Sprockets. Sprookets shall be of the split type and be constructed of cast iron. The cast iron shall be cast with a chilled rim to produce a surface hardness of' not less than 425 Brinnell, or shall be. class 40 cast iron conforming to ASTM A48 with a hardness of not less than 425 Brinnell. The sprockets shall be keyed to steel shafts which shall in turn be keyed to gearboxes or other driving mechanisms. The sprockets shall be assembled with the use of not less than four bolts per sprocket.

3.15.2.4 Idler rollers. The bucket line shall be provided with at least two idler rollers. The idler rollers shall be surface hardened to a Brinnell hardness of 400 to 450. Idler rollers shall be double flanged and secured to a steel shaft by means of key or set screw.

3.15.2.5 Takeup adjustment. A takeup adjustmentt of not less than 6 inches shall be provided. The takeup mechanism shall consist of two acme thread arrangements (one per side)* Takeup adjustments shall be supplied with both adjustment nuts and locknuts.

3.15.3 Power transmission. Power shall be transmitted to a gearbox on the elevator by means of a mechanical drive line from a outside power source.

3.15.4 Enclousure. The foot and upper run of the elevator shall be completely open. The return run shall be enclosed with removable, bolt-on housings extending to approximately 5 feet from ground level when the cold-aggregate elevator is in operating position. The enclosure shall be of steel sheet not less than 0.100 inch thick.

3.15.5 Discharge spout A' discharge spout shall be provided for compatible operation with the aggregate dryer drum feeders. The discharge spout shall be constructed of steel sheet not less than 0.100 inch thick.

3.16 Aggregate dryer. The aggregate dryer shall consist of a single rotating drum with feed or charging chute, oil burner, fuel pump, fuel piping, combustion chamber, discharge chute, gearboxes, drives, accessories, and instrumnts, and all other components necessary for the operation of the dryer as specified herein.

3.16.1 Capacity. The dryer shall have a capacity not less then that shown in table I when drying aggregate to a discharge temperature between 290°F. and 310° F. and with the dried aggregate having a maximum residual surface moisture content of 1/2 percent.

3.16.2 Drum. The dryer drum shall be a single cylinder constructed of welded steel plate conforming to ASTM A440 not less then 1/4 inch in thickness. The drum shall be provided with replaceable lifting flights bolted to the

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inner circumference which shall lift and drop the aggregate in a thin veil through the hot gases produced by the burner. A rotary elevator or other suitable means shall be provided at the burner end of the drum to discharge the dried aggregate into the boot of the hot-aggregate elevator or into the portable conveyor. The drum shall be fitted with not less than two alloy steel supporting tires bolted to the outer circumference. Each tire shall revolve on not less than two adjustable trunnions made of hardened cast steel. Means shall be provided for controlling the longitudinal movement of the drum along the longitudinal axis of the drum in both the uphill and downhill direction. The drum shall be driven by a positive-drive arrangement. If a sprocket and chain drive is employed, an adjustable, continuous oiling system shall be provided. All bearings used in the driving mechanism shall be of the antifriction type. The charging end of the dryer drum shall include a flanged exhaust gas duct connection to which the fines collector duct shall be attached.

3.16.3 Dryer drum feeder. The dryer drum feeder shall be of the reciprocating-feeder or positive-feeder chute type* Means shall be provided to eliminate draft losses. The feeder clute shall receive aggregate from the cold-aggregate elevator. Aggregate shall be fed continuoualy into the dryer drum without clogging. All drive Mechanisms necessary for the operation of the feeder shall be provided. The feeder assembly shall be rigidly supported by steel framework attached to the dryer trailer chassis. The framework shall include guides, supports, braces, and connections required for use in conjuction with the erection of and operation with the cold-aggregate elevator.

3.16.4 Dryer burner and fuel system. The dryer fuel system shall include burner, fuel pipings fittings, valves, fuel pump, fuel meter, strainers, preheater, preheater temperature control, relief valve, pressure gages, thermometers, and other items which may be required to provide a complete operating dryer fuel system.

3.16.4.1 Burner. The dryer drum burner shall be low-pressure, air-atomizing, and oil-burning. It shall burn number 1, 2, 3, and 4 fuel oil without preheating, and number 5 and 6 when preheaded. The burner shall have a capacity of not less than 300 gph. The burner shall operate in conjunction with the fines collector blower.

3.16.4.2 Fuel piping and pipe fittings. l Fuel pipe shall conform to WW-P-406. Pipe fittings shall be malleable iron. Unions shall ‘be installed where required for disassembly of piping, valves, and accessories. Unions shall conform to WW-U-531. Fuel piping shall be jacketed where required for hot-oil heating.

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3.16.4.3 Pipe supports. All fuel and hot-oil piping shall be provided with hangers or clamps where required to rigidly support the piping.

3.16.4.4 Valves. A control valve shall be installed in the fuel supply line to the butner i. a location accessible to the operator. The control valve shall provide an infinitely variable range of adjustment and shall maintain a steady flow of fuel at any setting. The control valve shall include a calibrated dial and indicating pointer to facilitate burner operating procedures. A gate-type, shutoff valve shall be installed in the fuel supply piping between the control valve and the fuel pump. The gate valve shall be of the same size as the piping in which it is installed and shall conform to WW-V-54. A relief valve shall be provided and installed in the fuel supply line to the burner. The relief valve shall have a capacity not less than the capacity of the fuel pump. A fuel-tank return line shall be connected to the discharge of the relief valve. All low points in the fuel-oil system shall be provided with drains to permit complete drainage of the piping.

3.16.4.5 fuel pump. The fuel pump shall be of the positive-displacement type for pumping the fuels specified for the burner, and shall have a capacity for not less than 300 gph. This capacity shall be developed when pumping against a discharge pressure not less than the combined friction loss created by the burner and fuelf-supply system. The pump shall be self-priming. The pump shall be provided with a positive-drive mechanism. The pump shall be permanently mounted on the dryer frame. ."

3.16.4.6 Strainers. A duplex oil strainer shall be provided and installed in the fuel-supply line in such a manner as to permit cleaning of one strainer while the other is in operation. Each strainer shall have the capacity to remove all foreign matter which might foul the burner while passing not less than 300 gph of oil of the grades specified for the burner.

3.16.4.7 Fuel meter. A flowmeter shall be provided and installed in the burner fuel-supply line. The fuel meter shall operate with the fuels specified for the burner, shall be of the recording type, and shall be provided with a glass-covered recording dial and hinged metal cover. The meter shall indicate the amount of fuel used in increments not greater than 1 gallon. The meter shall have a recording range of 0 to 99,999 gallons.

3.16.4.8 Fuel-oil preheater. A fuel-oil preheater shall be provided and shall be capable of heating not less than 300 gph of fuel oil from 100°F. to 200" F. The heating medium shall be oil conforming to MIL-L-22851, type III. The preheater shall be permanently mounted on the dryer frame.

3.16.4.9 Preheater temperature controller. An automatic temperature controller shall be provided for the fuel-oil preheater. The controller shall

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be equipped with a sensing element located in the fuel-oil outlet from the preheater, and with a manually controlled temperature selector and calibrated dial. The controller shall automatically maintain the fuel-oil outlet temperature within plus or minus 2°F. of any selected temperature in the range of 150° to 220°F.

3.16.4.10 Pressure gage. An oil-pressure gage shall be installed in the burner fuel-supply line. The gage shall be of the Bourdon type, shall operate with the fuel oils specified for the burner, shall have a heavy-gage metal case and a glass-covered indicating dial with a range of zero to 100 psig, and shall be provided with a shutoff cock or valve. The dial shall be not less than 3 inches in diameter and shall be graduated in increments of not snore than 5 psi.

3.16.4.11 Thermometers. Thermometers shall be installed in both the fuel-oil and hot-oil inlets and outlets from the preheater. The thermometers shall have scales not less than 6 inches long. Thermometers installed in the fuel-oil lines shall have 8 range of 40° F. to 275° F. Thermometers installed in the hot-oil lines shall have a range of 100° F. to 500° F. Thermometers. shall be installed so as to be visible for taking readings and accessible for replacement. The thermometer shall be shielded to prevent damage.

3.16.5 Hot-oil jumper lines. Flexible piping for hot-oil heating jumper Zincs to piping jackets and to fuel-oil preheater shall be provided.

3.16.6 Combustion chamber and ignition cone. The combustion chamber shall consist of a steel cylinder lined with firebrick conforming to HH-B-671, class 4, or refractory conforming to HH-R-191, class 1. The ignition cone shall consist of a steel cone lined with refractory conforming to HH-R-191, class 1. The combustion chamber and ignition cone shall be mounted at the discretion of the supplier.

3.16.7 Discharge elevator and discharge chute. Means shall be provided at the burner end of the dryer drum to discharge the dried aggregate into a discharge chute, which will in turn direct the aggregate into the boot of the hot-aggregate elevator or the portable conveyor. The elevator discharge shall be of such a height as to eliminate the need for a bat-aggregate elevator pit. A well for the sensing element of' the temperature indicator shall be installed in the discharge chute in a location which will result in a representative temperature of the discharge aggregate.

3.16.8 Temperature indicator. A pyrometer shall be provided to indicate the temperature of the aggregate as it is discharged from the dryer. The indicator shall have a range of 100° F. to 500° F. graduated in increments of not more than 5 degrees, and shall be provided with a weather-protecting enclosure. The sensing element shall consist of a sheathed thermocouple with protected cable for installation in the well provided in the discharge chute.

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3.16.9 Level inticator. A level indicator shall be provided and shall be mounted on the dryer assembly in such a manner as to indicate the angle of the longitudinal axis of the dryer drum drove the horizontal The indicator shall include a graduated scale in increments not greater than 1 degree, with 8 range of zero to 10 degrees. The level indicator shall be provided with a protective cover to prevent damage from westher and from falling aggregate. The indicator shall be mounted in a location where it will be visible and legible to the operator.

3.17 Fines (dust) collector. The fines collector shall consist of an ' exhaust fan, steel stack, centrifugal-type cyclone separators, screw conveyor, blower, metal duct, all necessary drives, gearboxes, clutches, controls, instruments, accessories, and other components necessary for the operation of the plant.

3.17.1 Exhaust fan. The exhaust fan shall be of a type for the intended. application and shall provide sufficient draft through the dryer and fines collector to result in efficient burner and separator operation. The capacity shall be not less than 24,000 cfm at the static pressure developed within the collector. The fan wheel and drive pulley or sprocket shall be keyed to the shaft. The exhaust fan shall have a single inlet to which the duct from the fines collector shall be attached, and a flanged discharge for connection to the stack. The fan shall be provided with a structural steel base which shall be bolted to the semitrailer chassis. Means for adjusting tension on the drive shall be provided. The base and fan housing shall rigidly support the weight of the steel stack. The fan housing shall be provided with inspection and cleanout plates.

3.17.2 Stack. A removable stack of a size compatible with the exhaust fan and extending not than 16 feet above grade shall be provided and shall be flange mounted on the discharge of the exhaust fan. The stack shall be constructed of steel not less than 0.137 inch thick.

3.17.3 Separators. The separators shall be of the centrifugal cyclone type. shall be of welded sheet steel or cast iron of not less than 0.1345 inch nominal thickness, and shall be complete with all necessary flanges for inlet and outlet connections l The capacity shall be such as to be compatible with the exhaust fan.

3.17.4 screw conveyor. A screw conveyor shall be attached to the bottom fines collector and shall convey the fines from the collector to the boot of the hot-aggregate elevator or portable conveyor. A combination screw convesyor and dusttight, enclosed-rubber-belt conveyor may also be used to convey the fines to the hot-aggregate elevator or portable conveyor t The conveyor tube shall be provided with inspection plates and cleanouts end with a discharge opening at the end. The conveyor may be constructed in sections securely bolted together l

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3.17.5 Blower. A centrifugal-type blower shall be provided to supply air for atomization to the burner. The blower shall have a centrifugal-type wheel, steel shaft and housing, and drive. The blower shall have a capacity to provide a sufficient quantity of air to the burner to assure efficient operation of the burner at maximum burner capacity. The blower shall be provided with not less than two antifriction-type bearings conforming to FF-B-185. The blower intake shall be provided with an expanded-metal or wire-mesh screen. The blower shall be rigidly mounted in a location at the discretion of the supplier. Means for adjusting drive tension shall be provided.

3.27.6. Ductwork. All metal ductwork and fittings required to connect the exhaust-gas outlet of the dryer drum to the inlet of the fines collector, to connect the outlet from the fines collector to the exhaust fan, and to connect the discharge of the blower to the atomizing air supply of the oil burner shall be provided. All necessary draw bands, gaskets, and flanged connections shall be provided to result in a tight duct system. Ducts shall be provided with structural steel supports to assure a sturdy installation. All ducts shall be removable for transport. A damper shall be provided in the duct between the fines collector and the exhaust fan to control draft.

3.18 Hot-aggregate elevators* Two hot-aggregate elevators shall be provided and shall conform to 3.25, except that each shall be totally enclosed and shall be provided with intake ports, flexible dusttight spout connections, and removable pneumatic-tired dollies and towing attachments unless permanently mounted with other components l

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3.19 Grradation-control unit. The gradation-control unit shall consist of a vibrating screen, a four-compartment bin with control gates, bin-level indicators, apron feeder or feeders, test boxes, chutes, housings, ladders, platforms, guards, and all other components required for the operation of the unit as specified herein.

3.19.1 Capacity. The 'gradation-control unit shall have a capacity to screen and feed not less than 150 tons per hour of either limestone or gravel in accordance with table I.

3.19.2 Vibrating screen. A horizontal or inclined vibrating-screen assembly with interchangeable wire screen cloths of the sizes specified herein shall be provided. The vibrating-screen assembly shall grade the aggregate into four uniform sizes and scalp off oversize material. The screen shall discharge each of the four sizes of aggregate into a separate compartment with the smallest aggregate being discharged into the largest (fine sand) compartment. The top deck shall be provided with an overflow chute to scalp off oversize aggregate. The screen shall be equipped with screen cloths of the following sizes:

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l-5/8-inch square opening
l-1/8-inch square opening
9/16-inch square opening
1/4-inch square opening

The screen shall consist of a stationary frame upon which is mounted a vibrating frame with the cloths specified above and with a vibrating mechanism.

3.19.2.1 Stationary frame. The stationary frame of the screen shall be constructed of structural steel members or steel castings cross-braced to prevent distortion under maximum loads and operating conditions as specified herein.

3.19.2.2 'Vibrating frame. The vibrating frame of the screen assembly shall be constructed of steel plates on structural steel members of rigid construction with sufficient crossmembers to securely hold the width to insure fitting of screen cloths. The vibrating frame shall be provided with means for holding the replaceable screen cloth securely in place to prevent movement of the cloth during full-load operation. A series of longitudinal support bars shall be provided which shall create a crown of not less than 1/2 inch in the screen cloth. Provisions shall be made for changing the cloth by using common tools. The frame shall be balanced to provide an even flow of material across the full width of the cloth.

3.19.2.3 Screen cloth.

3.19.2.3.1 Material. All screen cloth shall be made of crimping grade, oil-tempered wire conforming to NSGA Circular No. 80, table 3, grade B, carbon range 0.55 to 0.70, drawn and hardened as specified.

3.1%2.3.2 Wire size. Wire size shall conform to NSGA Circular No. 80, tole 7, as follows:

Clear Opening (rein) inch	Wire size
1-5/8	Standard heavy
1-3/8	"
1-1/8	"
7/8	"
9/16	
3/8	
1/4	Standard light
5/32	"
1/10	

Two screen cloths of each of the above sizes shall be furnished with each unit in addition to those specified in 3.19.2.

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3.19.2.4 Vibrating mechanism. A mechanical vibrating mechanism shall be provided for activating the vibrating-frame assembly. If material from the deck or decks of the screen could impinge on the activating mechanism, the mechanism shall be equipped with a replaceable, protective cover.

3.19.2.5 Shaft bearings. All shaft bearings on the vibrating screen shall 'be the double-row, self-alining, antifriction type. All bearings exposed to dust shall be provided with labyrinth-type grease and dust seals to prevent dust from entering the bearings. Drive shaft and bearings shall withstand the loads imposed under maximum operating conditions specified herein.

3.19.2.6 Screen housing. The entire screen assembly shall be housed with sheet steel of not less than 0.1345 inch nominal thickness to retain dust and heat. Access openings and covers shall 'be provided at all points requiring inspection or maintenance and shall permit individual screen cloths to be removed and replaced without disturbing the other screen cloths.

3.19.3 Screen flow-device(s). A quick-operating diversion device(s) shall be provided to divert the most coarse aggregate from designated compartment to the reject chute. The device(s) shall be operable from outside the unit and shall not necessitate removal of screen cloth.

3.19.4 Bin. The bin shall be fabricated of sheet steel of not less than 0.1345 inch nominal thickness braced and reinforced with structural steel shapes. The bin shall have a capacity of not less then 14 cubic yards of aggregate anti shall be divided into four compartments. Three of the compartments shall be of approximately equal capacity while the fourth, the fine-sand compartment, shall have a capacity of not less than 5 cubic yards.

3.19.4.1 Baffle plates. Baffle plates or other devices shall be provided and installed in the fine-sand compartment in such a manner as to provide positive control of fines flowing from the bin.

3.19.4.2 Calibrated control gates. Each of the four bin compartments shall be provided with a manually-operated calibrated gate with an adjustable opening from the closed position to an opening not less than 10 inches in height. Adjustments shall be possible in increments not greater then l/10 inch in height. A positive locking device and indicators for gate opening and capacity shall be provided for each gate. The locking devices and indicaters shall be located in a visible end accessible location.

3.19.4.3 Automatic shutoff. Each control gate shall be provided with an electrically-operated switch which will shut down operations when a gate fails to feed its preselected capacity. All necessary electrical connections, switches, wiring, and all other electrical devices required to connect to the mixer electrical system shall be provide.

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3.19.4.4 Telltales and overflows. Each of the four compartments shall be provided with an overflow chute and mechanical telltales to indicate and audibly warn when the level of' aggregate within the bin is insufficient for mixer operation.

3.19.5 Walkways and ladders. Steel walkways shall be provided at the top of the bin for access to bin compartments and screens for inspection, servicing, and maintenance- The walkways shall be provided with safety-tread, hand rails, and ladders.

3.19.6 Apron feeders. An apron feeder consisting of one or more continuous steel apron feeders, drives, pulleys, frame support, discharge chute, and all other components and accessories required for the operation of the feeder shall be provided. The feeder shall receive material from the calibrated gates of the storage bins and shall feed into the discharge chute. Where more than one feeder is provided, they shall be interconnected with positive, mechanical drives. The apron feeder shall be provided with line shaft, universal joints and gearboxes, sprocket and chain, or other equally suitable drives. A quick-change method to provide for not less than three feeder speeds shall be furnished.

3.19.7 Revolution counter. A revolution counter shall be provided which shall indicate the revolutions of the apron feeder pulley shaft. The indicator shall show revolutions in increments of 1/10 and 1/100 of a revolution The indicator shall be not less than a five-digit indicator. The indicator shall be provided with a glass cover and a metal protective hood and shall be permanently mounted so as to be visible to the operator at all times.

3.19.8 Discharge and test chutes. The apron feeder shall be provided with a discharge arrangement which will allow the aggregate to be directed either to a discharge chute which feeds into either the boot of the hot-aggregate elevator or into test chutes which feed into sampling containers. Test chutes shall feed material from each of the four bin compartments into separate sampling containers. The diverter shall be movable from one position to the other by means of a control lever or handle located in an unobstructed exterior position.

3.19.9 Sampling containers. Four sampling containers, each with a capacity of not less than 2 cubic feet of aggregate, shall be provided. The containers shall be constructed of steel. Each container shall be equipped with wheels or dollies supported by rails or platforms for positioning and removal. The containers shall be permanently stamped or marked "No. l," “No. 2," "No. 3," and "No. 4," and each shall be provided with not less than two heavy-duty handles securely attached to the container.

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3.19.10 Platform scale. A platform scale for weighing the sampling containers shall be provided. The scale shall have a capacity of not less than 300 pounds and shall conform to NBS Handbook H44.

3.20 Fines feeder. The fines feeder shall consist of a hopper, screw conveyor, drives, supporting frame, and all accessories or other components required for the operation of the fines feeder as specified herein. The fines feeder shall be either ground mounted or integral mounted with other components.

3.20.1 Capacity. The fines feeder shall have a capacity to feed not less than 4 cubic feet per minute of granular mineral filler.

3.20.2 Hopper, The hopper shall be constructed of sheet steel of not less than 0.1345 inch nominal thickness and shall have a capacity of not less than 300 pounds of granular mineral filler. The maximum height of the hopper shall be not greater than 7 feet. The hopper shall feed into a screw conveyor.

3.20.3 Screw conveyor. A screw conveyor or other means shall be provided under the hopper to accurately proportion the fines material within a range of 1/2 to 4 cubic feet per minute into the plant. Screw shafts shall be either solid steel or steel pipe and shall be provided with antifriction bearings. Screw flighting shall be steel not less than 3/32 inch thick at the outer edge.

3.20.4 Drive. The fines feeder shall be provided with shafts, universal joints and gearboxes, chain and sprocket, or other equally suitable drives. A jaw-type clutch shall be provided to disengage the feeder operation.

3.21 Portable conveyor. The portable conveyor shall conform to MIL-C-392, except that the length shall be as required to deliver aggregate into the mixer hopper as specified in 3.1 without exceeding the maximum slope as specified in MIL-C-392 when the mixer is supported by the operating jacks as specified in 3.9.5.

3.21.1 Gasoline engine. The gasoline engine shall conform to MIL-E-11275, group II. The following accessories, conforming MIL-E-112725, shall-be furnished:

(a) A 15-gallon fuel tank with fuel tank cap gasket, filler neck, and filler neck cap conforming to MS35643, M535644, and MS35645.
(b) Manual cranking system.
(c) 24-volt electric cranking system with 18-ampere charging generator and regulator, with batteries conforming to MS35000-1 and battery box, and without charging receptacle.
(d) Instrument panel with battery-generator indicator, fuel quantity indicator, hand choke control, starter control, high temperature control of engine shutoff type, and electric hourmeter conforming to MIL-M-3971/1-1.

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3.21.2 Clutch and clutch control. The engine shall be equipped with a hand-operated, over-center, engagement-type friction clutch with a rated capacity of not less than 150 percent of maximum engine torque. The clutch lining shall conform to KKK-L-370, type I or II. A hand control shall be provided for operation of the clutch by an operator standing on the ground.

3.22 Mixer. The asphalt mixer shall consist of a hopper and apron feeder if necessary, water pump transfer pump, metering pump, spray chamber, pug-mill, discharge chutes, and all other accessories, piping, and components necessary for the operation of the unit as specified herein. The mixer shall be supported to provide a minimum height of 8 feet between the discharge hopper and the ground when installed on level ground. The hopper and feeder maybe mounted on a cart-type trailer.

3.22.1 Capacity. The mixer shall have a capacity of not less than that specified in 3.1.1.

3.22.2 Intake hopper. When specified, an intake hopper of not less than. 8 cubic yards shall be provided to receive aggregate from the hot-aggregate elevator or the portable conveyor. The hopper shall be constructed of sheet steel not less than 0.100 inch in thickness. The hopper shall be equipped with a calibrated feed gate that can be adjusted to any position from closed to at least a 10-inch-high opening in increments of 1/16 inch, and shall be provided with a position indicator and positive locking device. The hopper shall be provided with a device to indicate when sufficient aggregate is in . the hopper for mixer operation.

3.22.3 Apron feeder. The intake hopper, when furnished, shall be provided with an apron feeder of sufficient capacity to supply aggregate to the pugmill during maximum rated mixer operations. The apron feeder shall be equipped. with flights and side rails to prevent leakage ad spillage of material. All parts of the feeder shall be accessible for removal of oversized or foreign material and for cleaning. The feeder shall be provided with all necessary drive for the feeder shall be connected to the asphalt metering pump drive 80 as to provide, automatically and accurately, a positive and continuous proportional delivery of aggregate and asphalt. A clutch shall be provided in the drive mechanism to allow disengagement of the feeder when it is desired to run the metering pump alone. The control lever for the clutch shall be located on the operator's platform. The apron feeder shall be equipped with a five-digit revolution counter to indicate the revolutions of the feeder foot shaft. The counter shall conform to 3.19.7.

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3.22.4 Water-supply system. The mixer shall be provided with a separate water-supply system for supplying water to the pugmill. The water-supply system shall be complete with water pump, water metier, spray bass, pipe, pipe fittings, valves, strainers, and all other components rewired to pro-tide a complete system, when supplied with water from a separate supply tank.

3.22.4.l Water pump A single-suction, single-stage, centrifugal water pump shall be provided for supplying water to the water spray bars. The pump shall have a capacity of not less than 85 gallons per minute (gpm) when pumping against a 20-foot head or against the friction head created by the installed spray bars and piping systems whichever is greater. All necessary drives shall be provided and shall 'be installed in a location such that the drive will be accessible for inspection, maintenance, repairs, and for removal when the water system is not to be operated.

3.22.4.2 Water meter. A water meter of the indicating type with a capacity of not less than 99,999 gallons in increments of not greater than 2 gallons shall be installed in the water supply piping to the spray bars. The meter shall be provided with an indicating dial. The indicating dial face shall have a glass cover and a protective lid of heavy metal. The water meter shall be equipped with a built-in strainer on the inlet side of the meter. The water meter shall be located so that the dial will be visible and legible to an operator on the operator's platform.

3.22.4.3 Water spray bars. Water spray bars shall consist of standard weight black pipe with holes drilled in one side. Spray bars shall be of such number and installed in such locations as to result in even distribution and thorough wetting of the aggregate in the pugmill and so that the spray can be directed to the sides of the pugmill if the material tends to build up in that area.

3.22.4.4 Water pipe, pipe fittings, and valves. Water pipe shall be standard weight black steel pipe, and shall extend from the mixer external supply connector to the water pump suction and from the pump discharge to the water spray bars. All fittings shall be standard weight malleable iron or cast iron. Unions shall be provided for removal of the water pump, water meter, and other equipment or piping, where required, for removal or dis-assembly of the piping system. A Y-type strainer of the same size as the pipe in which it is installed shall 'be provided and located between the pump dis-charge and the water meter. A cutoff valve shall be provided in the piping, end globe-type control valves shall be provided for each bank of sprays.

3.22.5 Transfer pump. The transfere pump shall be the positive-displacement, all-iron-body , jacketed type to pump bitumen from an external source. The pump shall have a capacity of not less than 80 gpm at a 10-foot head when

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pumping bitumen of penetration grade 85 to 100 at a temperature of 250° to 300° F. Penetration grades shall be as determined by ASTM D5. Both the pump and pump jacket shall be provided with a drain cock and an air-vent cock. The pump shall be provided with a shear pin safety device to prevent overloading the pump. The pump shall be provided with a drive, including a positive-engagement-type clutch, with control lever located on the operator's platform.

3.22.6 Hot-oil-heating piping l Hot-oil-heating piping shall be standard weight black steel pipe or seamless steel tubing, and shall extend from the external supply connections to the heating jackets of the transfer pump, metering pump, pugmill, bitumen piping, strainer, and spray bars. Pipe fittings shall be malleable iron or cast iron for black steel pipe. Pipe fittings for steel tubing shall be of steel.

3.22.7 ExternaL supply connectors. External supply connectors 3 inches in size shall be provided for connecting the asphalt transfer piping and the water pump piping to-an external source. Each connector shall consist of a faucet and connector and dust cap conforming to MS39153.

3.22.8 Metering pump. A positive-displacement, variable-volume, jacketed metering pump shall be provided to supply bitumen to the bitumen spray bars. The pump shall have a controllable capacity range of not less than from 9 to 70 gpm in not less than 30 preselected rates with an accuracy of within plus or minus 1 percent when pumping bitumen with a penetration grade of 85 to 100 at a temperature between 250° and 300° F. through the installed spray-bar system. Two pumps maybe supplied to meet the volume and accuracy requirements. If two pumps are provided, they may be either interchangeable or both mounted on the mixer. The arrangement shall allow both pumps to operate simultaneously. The pump shall be provided with a positive mechanical drive which shall be interlocked with the apron feed drive of the gradation control unit and shall include a clutch and shearpin safety device. Suction and discharge connections shall be flanged.

3.22.9 Bitumen sump and strainers. A jacketed bitumen sump and a strainer shall be provided and installed in the suction line to the bitumen metering pump. The sump and strainer shall consist of a heavy cast-iron or welded steel jacketed sump and a strainer with stainless steel or bronze screen. The strainer shall be constructed so that the screen maybe removed and the sump can be drained without disconnecting the piping. The sump jacket shall be connected to the hot-oil heating pipe system.

3,22.10 Spray chamber. An enclosed spray chamber shall be provided at the charging end of the pugmill. The chamber shall house the bitumen spray bars and shall be constructed of sheet steel. The chamber shall be provided with access doors or removable panels to permit inspection, cleaning, and removal and replacement of spray nozzles.

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3.22.11 Bitumen spray bars. The bitumen spray bars shall consist of lengths of standard weight pipe in which not less than five removable nozzles are mounted. The spray bars shall be mounted in the spray chamber with the nozzles directing the bitumen toward the aggregate. The nuzzles shall produce a fan-shaped spray pattern with not less than 50 percent overlapping spray extending across the entire width of the pugmill spray chamber. The spray bars shall discharge not less than 70 gpm of bitumen at a penetration grade of 85 to 100, at a tempereture between 250* and 300° F. The spray bar shall be jacketed for heating with hot oil.

3.22.12 Pugmill. The pugmill shall consist of an enclosed steel chamber (not less than 8 feet long with a total volume of not less than 60 cubic feet) in which is mounted not less than two paddle shafts equipped with paddles for mixing aggregate and bitumen or water into a homogeneous mixture. The size of the pugmill shall be such as to produce the capacity specified in 3.1.1 of a mix with not less than a 90 percent Ross count as defined in ASPT, Volume 28. The mixing time shall not exceed 45 seconds. The paddles and shafts shall permit adjustment for Controlling mixing action. Paddle-arm arrangement shall be such that any one paddle arm may be removed, reversed, or replaced without disturbing any of the other paddle arms. The shafts shall be constructed of steel, and the paddle arms of high carbon or nickel alloy steel. The paddle tips shall be removable, reversible, adjustable, and constructed of abrasive-resisting nickel-alloy steel or iron castings. The paddle tips shall be arranged to mix and carry the material toward the discharge end of the pugmill. An adjustable gate shall be provided at the discharge end of the pugmill which, in conjunction with the paddle adjustments, will control the mixing time of the material in the pugmill. The adjustable gate shall be controlled by a handwheel or other equally suitable means, and shall have a positive-position locking device. The gate controller should be located in a position accessible to an operator on the operator's platform. The bottom, sides, and charging end of the pugmill shall be provided with replaceable, abrasion-resisting, nickel-alloy-steel plate or castings not less than 3/4 inch in thickness. The pugmill shall be jacketed for heating with hot oil. The paddle shafts shall be supported at each end with steel. supports and antifriction bearings located outside of the mixing chamber. The pugmill housing shall be constructed of steel and shalll have a hinged steel cover. The pugmill shall be provided with the necessary drives or gear trains to receive power from the diesel engine driving the mixer. The drive shall be provided with a clutch with control lever located on the operator's platform.

3.22.13 Accessories.

3.22.13.1 Exhaust stack. A removable exhaust stack not less than 36 square inches in cross-section area and not less than 4 feet high shall be provided

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for the removal of fumes from the pugmill. The exhaust stack shall be constructed of' steel not less than 1/8 inch in thickness and shall be capable of removal from and replacement on the pugmill with the tools provided with the plant.

3.22.13.2 Discharge hopper. A hydraulically-controlled discharge hopper shall be provided and shall be installed at the discharge end of the pugmill during plant operations. The hopper shall have a capacity of not less than 2,000 pounds. When the hopper gates are opened, the entire mass of mix shall drop without restriction into a truck. The hopper gates shall be operated by means of a hydraulic ram receiving power from a hydraulic pump mounted on the mixer. The control for the hopper gates shall be accessible to the operator from the operator's platform. The hopper shall be constructed of steel sheet not less than 0.100 inch thick.

3.22.14 Controls and instrument panel. All operating controls of the mixer shall be located on the operator's platform and shall include but shall not be limited to controls for the following: Diesel engine master clutch, hot elevator clutch, metering pump clutch, transfer pump clutch, pugmill clutch, water pump controls pugmill adjustable gate, pugmill discharge hopper gates, and diesel engine throttle. If the instrument panel for the diesel engine furnishing power to the mixer is not visible to the operator from the operator's platform, duplicate engine control devices of the warning-light type shall be furnished and mounted on the platform.

3.22.15 Operator's platform. A steel platform of expanded metal (unflattened) with all necessary frames, guard rails, steps, ladders, and handholds shall be provided and installed on the mixer. The expanded metal shall have openings not greater than 3/4 inch, shall be of sufficient thick ness, and shall be provided with steel bracing and supports so as to result in a sturdy assembly. The platform shall be located so as to afford access to all controls and inspection points necessary to enable this operator to control all mixer operations as specified herein.

3.23 Piping system and wood blocking. Piping system (such as pipe, pipe fittings, and values) required to assemble the mixing plant in all central plant arrangments specified in 3.1 and shown on the drawings furnished under 3.1.4 shall be furnished. Wood burning shall not be furnished, but the dimensions and quantities of wood blocking components shall be shown on the drawings. The quality of piping system components and materials shall be as shown on DL13216E9731, but quantities and dimensions shall be applicable to the configuration of the mixing plant furnished. The quanity of each piping system component furnished shall be equal to the number required to assemble the mixing plant in tha central. plant arrangement that requires the largest number of the component. The quantities furnished will therefore permit any of the specified central plant arrangements to be set up separately, but no two arrangements simultaneously.

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3.254. Electromagnetic interference. The electromagnetic interference characteristics of the asphalt plant shall conform to MIL-STD-461, class III C.

3.25 Lubrication. All surfaces requiring lubrication shall be provided with suitable means for lubricating.

3.25.1 Lubricants. The mixing plant shall operate as required when lubricated as specified herein.

3.25.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, and III. Fittings shall be located in a protected position and shall be accessible to a hand operated grease gun with a 10 inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners maybe removed top provide accessibility.

3.25.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.25.4 Enclosures with integral reservoir of lubricant. Enclosures such as gearcases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped wirh a drain plug conforming to MS35844 or MS49006, and with a means for filling the enclosure with lubricate. The drain plug shall be located so that removal of the plu will result in complete drainage of the lubricant from the enclosure. Intgral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibilty to the drain plug, the filling means, and the lubricant-level chacking device shall be obtained without the removal or adjustment of accessories or parts, except that panels and ‘plates equipped with hand-operable quick-disconnect fasteners may be removed.

3.25.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases confoming to MIL-G-10924 or MIL-G-23827 as applicable The mixing plant shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubtrication because military greases are not compatible with other greases. A tag shall be attached in a conspicuous place on each unit to indicate which military grease has been used.

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3.25.6 Oil lubrication. AU components shall be filled to the operating level with winter grades of' oils conforming to MIL-L-2104, MIL-L-2105, or MIL-L-45199 as specified for the application. A tag shall be attached in a conspicous place on each unit to indicate the temperature range and grade of lubricant used.

3.26 Fungus and moisture resistance* The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 except that 1 percent copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish shall be used in lieu of saltcylanilide, the specified fungitoxicant:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements hose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces which have been previoysly cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.27 Lifting attachments. Each unit listed in 3.1 and each sectionalized component (see 3.7.2 and 3.7.3) shall be provided with lifting attachments. The attachments shall enable the unit or component to be lifted in its normal traveling or operating position. The attachments shall be located so that not less than l-inch clearance is maintained between the slings and all exterior parts of the item being lifted. Spreader bars are permitted. Each attachment shall withstand not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the slings will converge not more than 19 feet above the lowest extremity of the load when such limitation is possible. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the eye shall be not less than 3 inches. Members to which attachments are fastened and the fasteners shall withstand all stresses imposed by the attachment. Where required, structural members shall be reinforced to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.28.

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3.28 Tiedown attachments. Each unit listed in 3.1 and each sectionalized component shall be provided with integral attachments to permit tiedown to the floor or deck of the transportation medium. Each attachment shall withstand, without damage to the plant, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/4-G upward, and 1-1/2-G sideward, where G equals cargo weight of the unit or component, as applicable. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.27.

3.29 Identification marking. The mixing plant shall be identified in accordance with MIL-STD-130. Identification Plates shall conform to MIL-P-514, type I, style 1, composition, of type I, grade A, class l material. The supplier shall furnish and mount the required plates with screws or bolts in a conspicuous protected location.

3.30 Shipping data plates. Shipping data plates shall conform MIL-P-514, type III, composition c, of type X, grade A, class 1 material and in addition, shall show the silhouette of each unit and sectionalized component in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.31 Instruction plates. Each mixing plant shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the mixing plant. The words “Warning” and “Caution” shall be red matching color No. 31136 of FED. STD. No. 595. Instruction plates shall conform to MIL-P-514, type III, composition C, of type X, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.32 Treatment and painting. The portions of the mixing plant and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Parts subject to temperatures of 450°F. or over shall be painted with paint conforming to MIL-F-14105. Unless otherwise specified, the final coat shall match color No. X24087 of FED. STD. No. 595.

3.33 Vehicle making. The marking of each trailer and semitrailer shall be in accordance with MIL-STD-642. The gross vehicle weight of each trailer and the weight of each sectionalized component shall be marked on the side in such a manner as to be discernible. The prescribed tire pressure shall be marked near the tires. Lifting and tiedown attachments shall be marked "LIFT HERE" or "TIEDOWN HERE" as applicable.

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3.34 Technical publications. Such technical publications as ‘are specified shall be furnished.

3.35 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.36 Toolbox. Each trailer and semitrailer shall be provided with a weatherproof toolbox made of steel not less than O.071 inch thick. The toolbox shall be provided with a lid having a continuous hinge along one edge. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Two hook-type tension fasteners with strikers shall be provided to keep the lid closed. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a padlock having a 3/8-inch shackle shall be provided for locking the toolbox. The toolbox shall be fastened to the trailer or semitrailer in a protected position. A drain hole with rubber plug shall be provided in the bottom of the toolbox. A metal compartment shall be provided within the toolbox for storage of the technical publications. The toolbox shall be of a size to hold the technical publications and tool specified.

3.37 Items to be furnished with the mixing plant. No shipment of mixing plants shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the mixing plant without such items has been received from the contracting officer.

3.38 Workmanship. All parts, components, and assemblies of the mixing plant including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scales flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.38.1 Steel fabrication. Metal used in the fabrication of the mixing plant shall be free from kinks and sharp bends. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting of steel may be employed instead of shearing or sawings. Burned surfaces of flame-cut steel shall be ground or machined sufficiently to remove ash and cooling checks. All burrs and rough edges shall be removed. Precautions shall be taken to avoid overheating, and heated metals shall be

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allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.38.2 Bolted connections. AU threaded parts shall be in accordance with Handbook H28. Washers and lockwashers or elastic stop nuts shall be provided for all bolts. Nuts and screws shall be tight. Each toothed lockwasher shall be attached to its bolt or screw.

3.38.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.38.4 Welders and welding.

3.38.4.1. Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME Boiler and Pressure Vessel Code.

3.38.4.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved 'by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection, Inspection shall be classified as follows:

Preproduction inspection (see 4.3).
b. Initial production inspection (see 4.4) l
(c) Quality conformance inspection (see 4.5).
d Inspection comparison (see 4.7).
e Inspection of preparation for delivery (see 4.8).

4*3 Preproduction inspection.

4.3.1 Engine performance. Prior to examination and test of the plant, the engines that will be installed in the preproduction model plant components shall be tested for conformacce to MIL_STD-1400, Test Methods Series 2000. Nonconformance to 3.13.1 or failure to meet the requirements of MIL-STD-1400 Test Methods Series 2000 shall constitute failure of this test.

4.3.2 Examination. The mixing plant shall be examined as specified' in 4.6.l. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The mixing plant shall be tested as specified in 4.6.2. Failure of any test shall be cause for **rejection.**

4.4 Initial production inspection. An initial production mixing plant(s) will be selected at random by the Goverment from the first

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five mixing plants being reduced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production mixing plant shall not exclude the remaining mixing plants from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of an initial production mixing plant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production mixing plant(s) and shall be cause for refusal by the Government to continue acceptance of production mixing plants until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on mixing plants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all mixing plants accepted prior to completion of initial production inspection exe simiarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each mixing plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each mixing plant shall be tested as specified in 4.6.2.9. Failure of the test shall be cause for rejection.

4.6 Inspection procedure l

4.6.1 Examination. The mixing plant shall be examined for the following defects:

101. Dimensions not as specified.
102. Fasteners not as specified*
103. Assembly not complete or not correct.
104. Parts or components missing or not as specified.
l05. Parts or components do not function.
106. Safety devices or guards missing or not as specified.

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107. Material not as specified.
108. Treatment and painting not as specified.
109. Lubrication devices not accessible or not as specified.
310* Identification or other marking, operating and safety instruction, or manuals missing.
111. Workmanship not as specified.
112. Electromagnetic interference suppression components and " network not identical with approved model.

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to being tested, the mixing plant shall be lubricated with oils and greases specified in 3 l 25. Oils shall be those designated for use in the ambient temperature at which tests are being performed During all plant tests, the engines shall be governed at normal operating speeds which shall not be greater than the highest continuous speed for which the engine has been qualified Wood blocking for tests shall be furnished by the supplier. Where individual components are tested with the use of power sources other than the mixing-pl=nt engines, the component will be operated at the normal plant operating speeds. As services and adjustments are performed, the mixing plant shall be checked for ease of maintenance. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.2 Hydrostatic l Subject the items shown in tible IV to the pressures specified therein for the time periods indicated. Any leakage resulting from these tests shall constitute failure of this 'test.

Table IV. Hydrostatic Test Pressures

Item	Pressure	Time
	psi	hr
Dryer-burner fuel piping, including preheater	150	4
All hot-oil-heating piping including pipe jacketing and heater	150	4
Mixer water piping	50	4
Asphalt piping	50	4

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4.6.2.3 Pump capacities. Run capacity tests on the pumps as shown in table V under the conditiona specified therein. Unless otherwise specified herein, pumps shall be tested in their assembled positions while being driven by the regalar mixing-plant drive systems at normal operating speeds. Capacities less than those specified in table V shall constitute failure of this test.

Table V. Pump Capacity Tests

Pump	Liquid	Temperature of liquid	Suction lift	Discharge head	Capacity
		° F.	feet		
Dryer fuel	Fuel oil, VV-F-815, grade F.S. No. 5.	95 to 105	10	*	300 gph
Mixer water	Water.	40 to 60	10	20	85 gph
Mixer transfer	Asphalt cement, penetration grade 85-100, ASTM D5.	250 to 300	10	10	80 gph
Mixer metering	Asphalt cement, penetration grade 85-100.	250 to 300	10	*	**10 gph 30 gph 50 gph 70 gph

* Not less than head created by the installed piping.
** Tests shall be made at preselected rates nearest those shown. Accuracy shall be within plus or minus 1 percent. Laboratory test by pump manufacturer is acceptable.

4.6.2.4 Cold-aggregate feeder hopper capacities. Fill the coarse-aggregate hopper and the fine-aggregate hopper, both with and without bin extensions, with aggregate. Then remove and weigh the aggregate. Capacities less than those specified in 3.14.2.1 and 3.14.2.2 shall constitute failure of this test l

4.6.2.5 Dryer fuel meter. Test the dryer fuel meter for accuracy by diverting the fuel from the burner to a container. Weigh the fuel after it has flowed into the container at a constant rate for 1 minute. Test at 1/4, 1/2, 3/4, and full capacity with number 2 grade oil without preheating and nuymber 5 grade oil preheated to 200° F. Failure of the meter to record properly the flow of fuel at any capacity shall constitute failure, of this test.

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4.6.2.6 Dryer, fuel line heating jacket. Supply the dryer fuel line heating jacket with a hot-oil-heating medium at a temperature not greater than 450° F. Discharge the fuel oil. Record the temperature of the fuel at entrance and at discharge for a period of not less than 2 hours. Failure to raise the fuel oil temperature not less than 25° F. when passing 300 gph fuel oil through the heating jacket shall constitute failure of this test.

4.6.2.7 Fuel oil preheater and preheater controller. Supply fuel oil conforming to VV-F0815, grade F.S. No.5, to the preheater at a temperature between 95° F. and 105° F. at the rate of 300 gph for a period of not less than 2 hours. The preheater and preheater controller should maintain a discharge fuel oil temperature of 200° F. when hot oil at a temperature of not greater than 450° F. is used as a heating medium. Failure of the preheater and preheater controller to maintain the specified fuel oil discharge temperature shall constitute failure of this test.

4.6.2.8 Dryer burner. The dryer burner test shall be conducted while the burner is installed in the dryer drum and is operating under normal plant operations* Operate the burner for not less than 8 hours. This test shall be conducted both with number 2 grade fuel oil without preheat and number 5 grade oil preheated to 200° F. Failure of the burner to operate with each fuel or the development of uneven flame configuration or hot spots within the dryer drum shall constitute failure of this test.

4.6.2.9 Compatability. Assemble and operate for not less than 1 hour the units and sections in each of the following three arrangements: Multiple aggregate mix with hot-aggregate elevator and gradation-control unit; single aggregate mix with hot-aggregate elevator; single aggregate mix with belt conveyor. These tests may be conducted either loaded or unloaded provided that opetating unloaded will cause no damage to any unit or to the plant. The piping equipment and wood blocking required to assemble the plant in the above specified three arrangements shall be checked for conformance to the erection drawings furnished by the supplier. Failure of a unit or component to assemble and operate without modification shall constitute failure of this test.

4.6.2.10 Plant capacity. Operate the mixing plant with gradation" control unit for not less than 1 hour of continous operation using either limestone or a gravel with an initial moisture content of not less than 5 percent as specified in table 1. Repeat the test using aggregate with a minimum initial moisture content of not less than

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9 percent. Failure to produce high-type asphaltic concrete mix at the capacities shown in table Ior nonconformance to 3.23 shall constitute failure of this tests

4.6.2.11 Individual capacity. Operate the individual units specified in 3.14 through 3.32 for a period of not less than 1 hour. Determine **the capacity of each unit. Individual unit tests may be made concurrently with other tests. Individual capacity tests, except for units specified in 3.17, 3.21, and 3.22, may be made using power sources other than normal asphalt plant power when in accordance with 4.6.2.1. Failure of any unit to meet minimum specified capacities shall constitute failure of this test.**

4.6.2.12 Vehicle performance test. Tow each trailer using a Government-loanded, 5-ton, 6 by 6, M52 truck-tractor (see 6.4) over highways for not **less than 100 miles. Tow each trailer over unimproved roads through rolling and hilly terrain for not less than 25 miles. During or after the cumulative 125 miles on each trailer, trailing, turning, and braking ability shall be observed. Inability of any trailer to operate, trail, turn, retard, and stop as specified; or damage to or permanent deformation of any part of any trailer shall constitute failure of this test.**

4.6.2.13 Tandem-axle-assembly articulation. Tow the unit over 8-inch-high obstacles such that all combinations of wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or later interference, and, after returning the tandems to normal, measure the tendems for evidence of mislinement.

4.6.2.14 Rail and air transport. Sectionalize each unit of the mixing plant for rail and for air transport. Then reassemble each unit. Nonconformance to 3.7.2 and 3.7.3 shall constitute failure of these tests. The test specified in 4.6.2.15 for lifting and tiedown attachments may be accomplished in conjunction with this test.

4.6.2.15 Tiedown and lifting attachments. Apply the static load as l ewlfled in 3.27 for lifting attachments and in 3.28 for tiedown attachments to each attachment when mounted on its structural member. Weld failure or permanent deformation of the attachment, fastenings, or any structural member shall constitute failure of this test*

4.6.2.16 Electromagnetic interference. The preproduction model shall be tested to determine conformance with 3.24. The supplier shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select mixing plants at any time during the contract production period and subject these plants to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine that the quality of the selected mixing plant is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Mixing plants will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production mixing plants.

4.7.1 Inspection failure. Failure of any mixing plant when examined and tested as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of mixing plants until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defeats shall be cause for re jection.

4.8.1.2 Test. The preproduction pack of crated components for level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. The inspection level shall be S-2 and the acceptable quality level (AQL) 6.5 percent defective.

4.8.2.3 Examination. Sample selected in accordance with 4.8.2.2 be examined for the following defects. Presence of one or more def shall be cause for rejection

333. , Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
114. Sectionalization and disaasembly not as specified for level A or B.
115. Parts not matchmarked to facilitate reassembly for level A or B.
116. Cloth tags for matchmarking and lubrication information not waterpoofed for level A.
117. Marchmarking and lubrication information tags not attended for level A or B.
118. Unpainted exterior metal surfaces not coated with preservative for level A or B.
119. Exposed exterior machined surfaces not coated with preservative, wrapped, or covered with barrier material, and barrier material not l mred as specified for level A or B.
120. Trailer chassis not preserved in accordance with the referenced document for level A or B.
121. Engines and engine accessories not preserved and packaged in accordance with the referenced document.
122. Clutches not preserved as specified for level A or B.
123. Gears not preserved as specified for level A or B.
124. Drive chains not preserved as specified for level A or B.
125. Shrifts and universal joints not coated with preservative a specified for level A or B.
126. Drive belts not removed or released from tension for level A or B.
127. Unpainted pulley grooves not coated with primer as specified for level A.
128. Belt fasteners not coated with varnish as specified for level A.
129. Conveyor belt not released from tension for level A or B.
130. Conveyor belt when removed not compactly rolled and secured as specified for level A or B.
131. Wire rope cables not coated with preservative, and loose cables not coiled and secured as specified for level A or B.

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132. Heating unit not preserved and openings not sealed as specified for level A or B.
133. Temperature indicator not preserved as specified for level A or B.
134. Level indicator not preserved as specified for level A or B.
135. Screens not preserved as specified for level A or B.
336. Platform scales not disassembled, preserved, and wrapped as specified for level A or B.
137. Hydraulic systems not preseved as specified for level A or B.
138. Pumps not preserved and openings not sealed as specified for level A Or B.
139. Pugmill not preserved, covers not closed and secured, and openings into the chamber not sealed as specified far level A
140. Repair parts not preserved In accordance with the referenced document.
141. Miantenance tools not preserved and packaged in accordance with referenced document for level A or B.
142. Toolbox lids not secured to prevent pilferage for level A or C.
143. Disassembled components or intermediate containers placed on the units in a manner that would increase cubage, or interfere with lifting or mobility for level A.
144. Blocking, bracing, and anchoring of crated components not in accordance with the referenced document for level A.
145. Depreservation guides not prepared, placed in waterproof envelopes, and securd to the basic unit for which intended.
146* Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5*1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified to prove, prior to starting production packaging and packing, that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements or this specification (see 6.5). Examination and tests shall be those specified herein.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified.

5.2.1 Level A.

5.2.1.1 Sectionalization and dissassembly. The plant shall be sectionalized to meet the requirments of 3.7.2 and 3.7.3, as applicable. In addition;

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components of units and sections vulnerable to damage, pilferage, and loss, and protruding parts that increase cubage and that can be removed and reassembled without special tools shall be removed. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss. Removed keys shall be secured in the keyway of the pulley or flywheel.

5.2.1,2" Matchmarking. To facilitate reassembly, removed components and sections shall be matched. Large sections and components shall be matchmarked by stenciled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-l2l or MIL-V-1174 or matchmarked with paint conforming to MIL-P-13086. Small components shall have matchmarking information on cloth tags Conforming to UU-T type A, or on metal tags. The cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to 'the preservation or to the components.

5.2.%.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces.

5.2.1.4.1 Unfinished surfaces. Components of the mixing plant such as tie rods, adjusting rods, clevises, coil springs, shafts, couplings, blowers, screw conveyors, sampling containers, gates, brackets, framing, leaf springs, linkages, ratchets, pawls, liner sections, exhaust stacks, grease fittings, washere, pins, bolts, nuts, or any other surface exposed by sectionalization or disassembly, or any other similar unpainted exterior metal surface not definitely specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. Unpainted matal identification and instruction plates shall be creed with varnish conforming to MIL-V-13811.

5.2.1.4.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to PPP-T-60, type I.

5.2.1.5 Trailer components. The components of the trailer and semitrailer chassis shall be preserved and packaged in accordance with MIL-STD-281, Level A (mobile).

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5.2.1.6 Engines. Engines and engine accessories shall be preserved and packaged in accordance with MIL-E-10062, level A.

5.2.1.7 Clutches.

5.2.1.7.1 Disc type. The inspection plate shall be removed, and with the clutch engaged, all accessible interior metal components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to a minimum. Clutch-control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, the spring-loaded-type clutches shall be secured in a partially disengaged position to eliminate contact between the disc facings and pressure plates. Snap~r-center and toggle-in-type clutes shall be completely disengaged. The clutch shall not be operated after application of the preservative.

5.2.1.7.2 Jaw type. Cluch jaws, shifter yokes, and machined surfaces of shafts shall be coated with type P-6 preservative. Clutch control machanisms enclosed with housings shall be fogged with type P-10, grade 2 preservative l Clutch control mechanisms not enclosed within housings shall be coated with type P-l preservative. The clutch shall not be operated after application of the preservative.

5.2.1.7.3 Band and shoe type. Clutch drum facings and control mechanisms enclosed in the housing shall be coated with primer conforming to TT-P-664. Clutch control mechanisms not enclosed shall be coated with type P-l preservation. The clutch shall not be operated after application of the preservative.

5.2.1.7.4 Cone type. The unlined cone and clutch control mechanism shall be coated with a thin film of primer conforming to TT-P-664. The clutch shall be disengaged to prevent the lining from adhering to the unlined cones. The clutch shall not be operated after application of the preservative.

5.2.1.8 Gears.

5.2.1.8.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.8.2 Enclosed gears. Gears operating on lubricating oil or lubricant shall be filled to the operating level with the approved

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lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable-type lubricant." Tags shall conform to UU-T-81, type A. Tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.9 Drive chains.

5.2.1.9.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to #e inner surfaces of the rollers, pins, and bushings. The excess preservation than be drained, and the entire chain shall be coated with type P-1 preservative. Removed chains shall be coiled, and the coil shall be secured with soft, annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade At class 2, secured with tape conforming 'to PPP-T-60, type I. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.9.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.8.2 for enclosed gears and gear housings.

5.2.1.9.3 Bucket-carry chains. Bucket-carry chains and attached buckets shall be coated with type P-1 preservative. Removed chains shall be coiled, and the coil shall be secured with zinc-coated strapping conforming to QQ-S-781 or QQ-S-790, size w applicable l

5.2.1.10 Shafts and universal joints. Exposed surfaces of shafts and universal joints shall be coated with type P-l preservative.

5.2.1.11 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.1.12 Conveyor belts and fasteners. Metal belt fasteners shall be coated with varnish conforming to MIL-V-1174 or TT-V-l21. The belt shall be removed or released from tension. Removed belts shall be compactly rolled, and the roll shall be secured with strapping conforming to QQ-S-781, type I, class B, size as applicable. The straps shall be placed circumferentially on the roll and the straps shall be tensioned so as not to damage the belt . Unpainted surfaces of the rollers shall be coated with primar Conforming to TT-P-664.

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5.2.1.13 Wire-rope cables. Wire-rope cables shall be coated with type P-1 preservative. After the preservative has dried, loose cables shall be individually coiled and the coils shall be secured with evenly spaced wire ties l

5.2.1.14 Drum and rotary elevator. All exterior unpainted metal surfaces of the drier drum rotary elevators drum tires, trunnion, thrust rollers, sprockets, and unpainted metal surfaces within the drum and the rotary elevator, including the lifting flights, shall be coated with type P-l preservative.

5.2.1.15 Heating unit.

5.2.1 .15.1 Combustion chamber and cone. All unpainted metal surfaces of the combustion chamber and cone, brackets, dampers, cover plates, and similar items shall be coated with type P-1 preservative. Openings into the combustion chamber and cone shall be covered with tape conforming to PPP-T-60, type I.

5.2.1 .15.2 Burner. The interior surfaces of the burner shall be coated with type P-10, grade 1 preservative, and all openings into the burner shall be sealed with tape conforming to PPP-T-60, type I. Unpainted exterior surfaces of the burner shall be coated with type P-1 preservative.

5.2.1 .15.3 Fuel tank, pump, preheater, piping, and valves. All interior surfaces of the fuel tank, pump, preheater, rotors, rotor shafts, vaues, vane slots, oil passages, piping, valves, springs, and similar parts shall be coated with type P-10, grade 1 preservative. The pump shall be actuated during application of the preservative to insure complete coverage of all interior parts of the system. Excess preservative shall be drained. All unpainted exterior surfaces of the fuel pump, preheater, piping, and valves shall be coated with type P-1 preservative. All openings to the interior of the system shall be sealed with tape conforming to PPP-T-60, type I.

5.2.1.15.4 Fuel-oil strainers. Each fuel-oil strainer shall be removed and cleaned, and the metal surfaces shall be coated with type **P-10, grade** 1 preservative and replaced.

5.2.1.16 Temperature indiactor. The temperature indicator shall be **removed and** preserved in accordance with MIL-P-116, method IIb. The

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inner container shall be a type CF, class domestic, variety SW, grade 125, style optional box and the outer container shall be a V3c, style optional box, both conforming to PPP-B-636.

5.2.1.17 Level indicator. Unpainted metal surfaces of the level indicator shall be coated with type P-10, grade 2 preservative. Moving parts shall be secured,. and the indicator shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, type I.

5.2.1.18 Screens. Unpainted surfaces of the screens shall be coated with type P-1 preservative. Screens not secured in the unit shall be bundled together by securing with flat steel strapping conforming to QQ-S-781, type I, class B, size as applicable.

5.2.1.19 Platform scales. The seales shall be disassembled. Surfaces of components requiring a contact preservative shall be coated with type P-7 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and the wrap shall be secured with tape conforming to PPP-T-60, type I. Components comprising each complete scale shall be intermediate packaged together in a container specified in 5.2.1.27.

5.2.1.20 Hydraulic systems. The fluid supply tank, pump, and piping shall be filled with hydraulic oil as recommended by the supplier. The hydraulic system shall be actuated to assure complete coverage of all interior surfaces and parts with the fluid. The ram shall be fully retracted. Unpainted surfaces of the the, tank, piping, prep, valves, hopper, control linkage, cable, and chain shall be coated with type P-1 preservative.

5.2.1.21 Pumps. Pumps shall be preserved as follows: While the pumps are being actuated, type P-l0, grade 2 preservative shall be applied through the pump intakes in sufficient quantity to assure coating of all interior surfaces all parts of the pump, piping, values, spraybars, jets, and nozzles. Excess preservative shall be drained. Openings into the spraybars, jets, and nozzles shall be saeled with tape conforming to PPP-T-60, type I.

5.2.%.22 Pugmill. Exterior unpainted surfaces of the pugmill shall be coated with type P-1 preservative. Interior surfaces of the mixing chamber, including the shafts and paddles, shall be coated with type P-10, grade 2 preservative, applied by fogging or spraying. The pugmill

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covers shall be closed and secured, and any openings into the chamber that will permit the entrance of snow or water shall be sealed with tape conforming to PET-MO, type I.

5.2.1.23 Repair The preservativew application critieria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts l

5.2.1.24 Maintenance tools. Maintenance tools shall be preserved in accordance with MIL-H-15424, level A and placed in the toolbox.

5.2.1.25 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1, and placed in the toolbox.

5.2.1.26 Other components. Other components not specifically mentioned herein shall be prose-d as specified fur similar components.

5.2.1.2'7 Intermediate packaging. Packaged and unpackaged components, accessories, and repair parts that cannot be secured to the basic units or sections shall be intermediate packaged in the toolboxes as space pemits or in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional.

5.2.2 Level B. The complete mixing plant shall be preserved and packaged as specified for level A, with the following exceptions l

5.2.2.1 Tags. Cloth tags specified in 5.2.1.2 and 5.2.1.8.2 shall not require waterproofing.

5.2.2.2 Trailer components. The components of the trailer and semitrailer chassis shall be preserved and packaged in accordance with MIL-STD-281, level B (mobile).

5.2.2.3 Exposed chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 or P-9 preservative shall be applied.

5.2.2.4 Drive belts. Drive belts shall be released from tension.

5.2.2.5 Conveyor belt fasteners. Belt fasteners are not required to be coated with varnish.

5.2.2.6 Temperature indicator, The temperature indicator shall be removed and preserved in accordance with MIL-P-116, method IA-14.

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The inner container shall be a type CF, class domestic, variety SW, grade 125, style optional box and the outer container shall be a class W5c, style optional box, both conforming to PPP-B-636.

5.2.2.7 Intermediate packaging. Intermediate packaging shall be as specified for level A, except boxes shall be domestic type or class as applicable.

5.2.3 Level C.

5.2.3.1 Repair parts, maintenance tools, and technical publications. Repair parts and technical publications shall be preserved and packaged as specified for level A. Maintenance tools shall not require preservation but shall be place in the toolbox as specified for level A.

5.2.3.2 Engines. Engines shall be preserved in accordance with MIL-E-10062, level C.

5.2.3.3 Other components. Preseration and packaging of all other components of the mixing plant shall afford protection against deterioration and damage from the supplier to the initial destination.

5.3. Packing.1 Packing shall be level A or C, as specified.

5.3.1 Level A. The mixing plant shall be prepared for mobile shipment. The toolbox lids shall be closed and secured to prevent pilferage. When sufficient space is available, packaged and unpackaged components required for each unit or section shall be located on the unit or section for which the components are intended. The components shall be placed and secured in any available space that will not increase cubage or interfere with lifting or mobility of the units. The component shall be secured to the units with zinc-coated strapping conforming to QQ-S-781 or QQ-S-790. Packaged and unpackaged components that cannot be secured to the units or to the sections shall be packed in crates conforming to PPP-C-650, style A, type optional; MIL-C-3774, nailed assembly, skid-type base; or MIL-C-104, type I, class 1 or 2, style A or B, as applicable. Blocking, bracing, and anchoring shall be in accordance with the applicable crate specification and with MIL-STD-1186. Crated components shall be shipped at the same time as the unboxed (mobile) units.

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5.3.2 Level C. The comoplete mixing plant shall be prepared for unboxed (mobile) shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rates. The toolbox lids shall be closed and secured in a manner to prevent pilferage. When efficient space is available, packaged and unpackaged components required for each unit or section shall be placed and secured on the unit or section for which intended in any available space that will not interfere with lifting or mobility of the units. Packaged and unpackaged components that cannot be secured to the units or sections shall be packed in crates complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules. Crated components shall be shipped at the same time as the unboxed (mobile) units.

5.4 Depreservation guide. Depreservation guides shall be prepared, placed is waterproof envelopes marked "Depreservation Guide," and secured in a conspicuous location on each basic unit. Unless otherwise specified (see 6.6), DA Form 2258 - "Depreservation Guide for Vehicles and Equipment" shall be used.

5.5 Marking. Packages, shipping containers, and unboxed (mobile) units shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The mixing plant is intended for use as a complete central plant (except for burner fuel storage, hot oil heating, asphalt melters, bulk bituminouns storage, and asphalt storage transfer pump facilities) for producing continuous mix, high-type asphaltic concrete mixes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame for submission of the preproduction model (see 3.2).
(c) When an initial production mixing plant is required and number of mixing plants to be furnished for initial production inspection, when applicable (see 3.3) .
d When mixer intake hopper is to be furnished (see 3.22.2).
e) When color other than as specified in 3.32 is required.
(f) Techincal publications required (see 3.34).
(g) Applicable list of repair parts and maintenance tools (see 3.35).
(h) Time frame for submission of preproduction pack (see 5.1).

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(i) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the mixing plant will be held in covered storage.
(j) When other than DA Form 2258 is to be used (see 5.4).

6.3 Preproduction model. Any changes or deviations from the preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish mixing plants Conforming 'to this specification.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 4.6.2.12. The M52 truck-tractor is one of the M39 series truck and can therefore be used for testing both semitrailer and cart-type trailer-mounted units.

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting aff'titer. Approval of' the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and the mixing plants in accordance with this specification.

6.6 " Depreservation guide The contracting officer should arrange to furnish DA Form 2258 when requested 'by the' supplier.

custodians:

Army - ME
Navy - YD
Air Force - 84

Review activity:

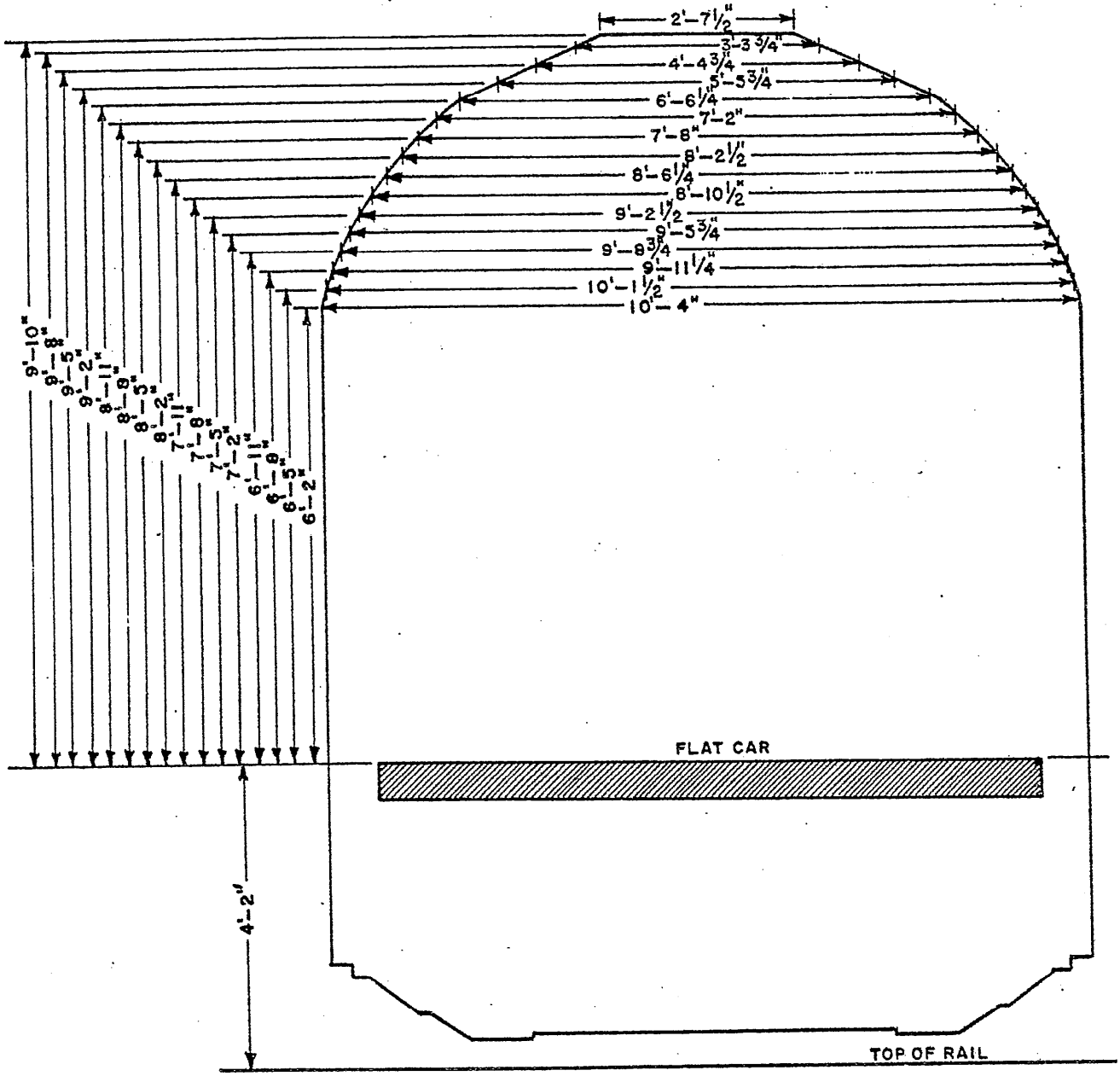
Army - EL

Preparing activity:

Army - ME

Project No. 3895-0166

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ITEM LENGTH-32'-10"

Figure 1 BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

A13123-1

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EQUIPMENT LOADING PLANNER
C-133A

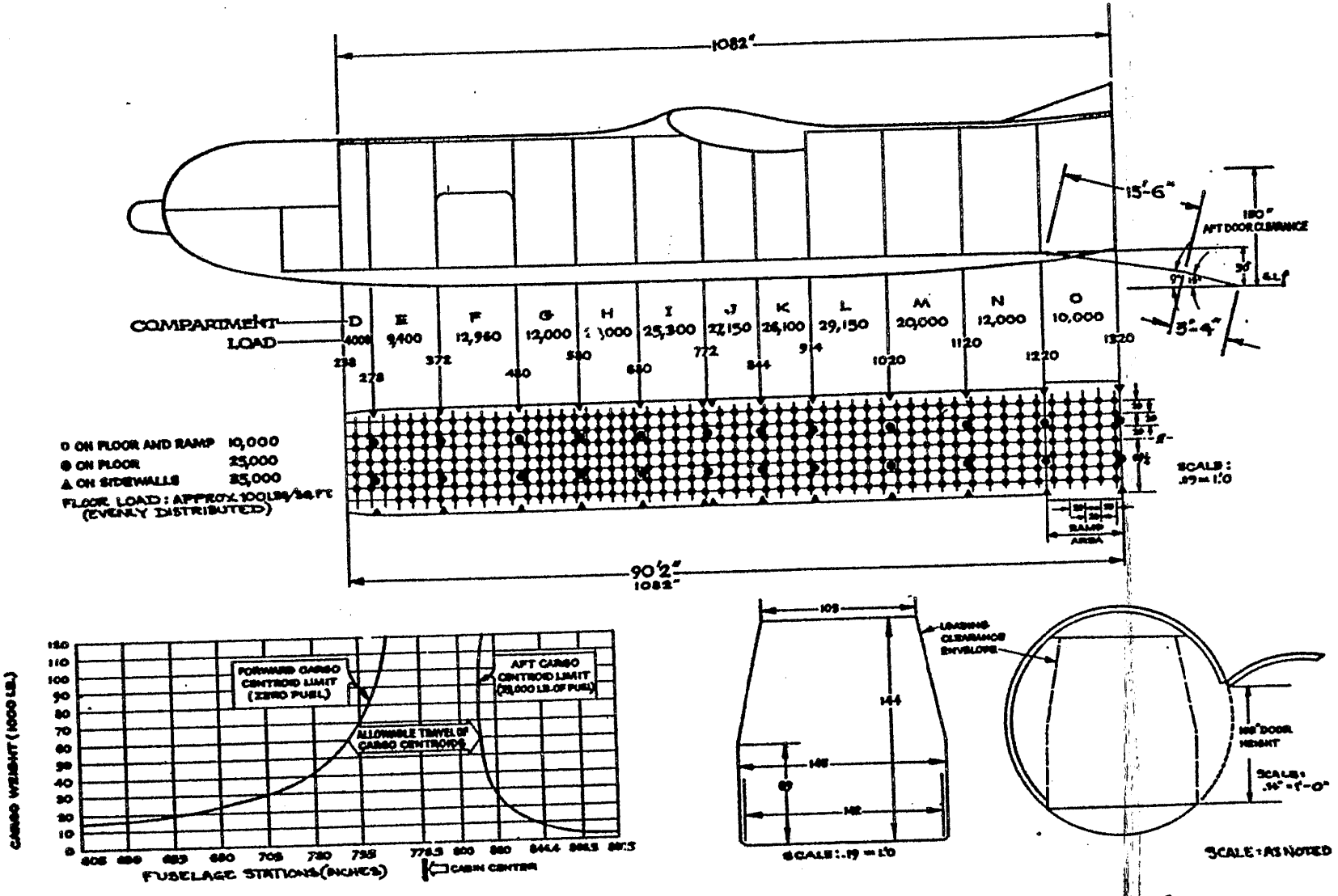
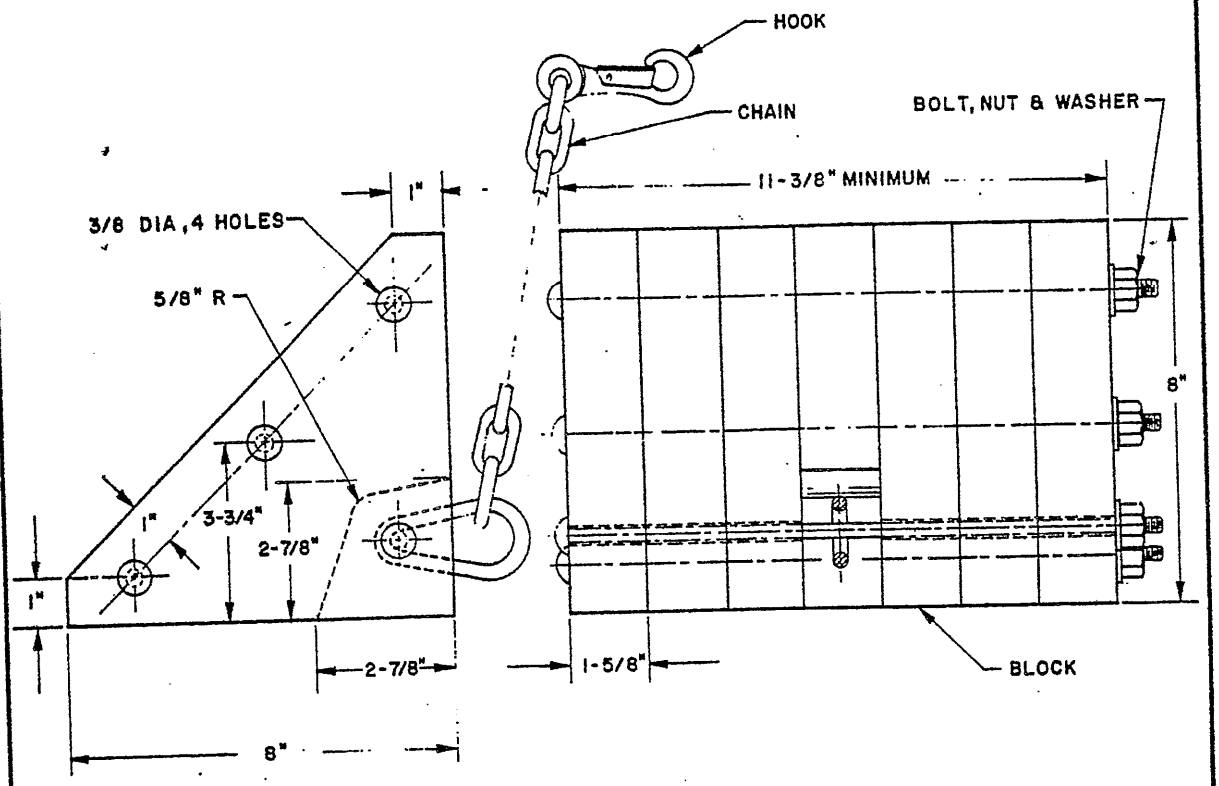


FIGURE 2.

T-969

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT/100 FT. = 70 LBS.

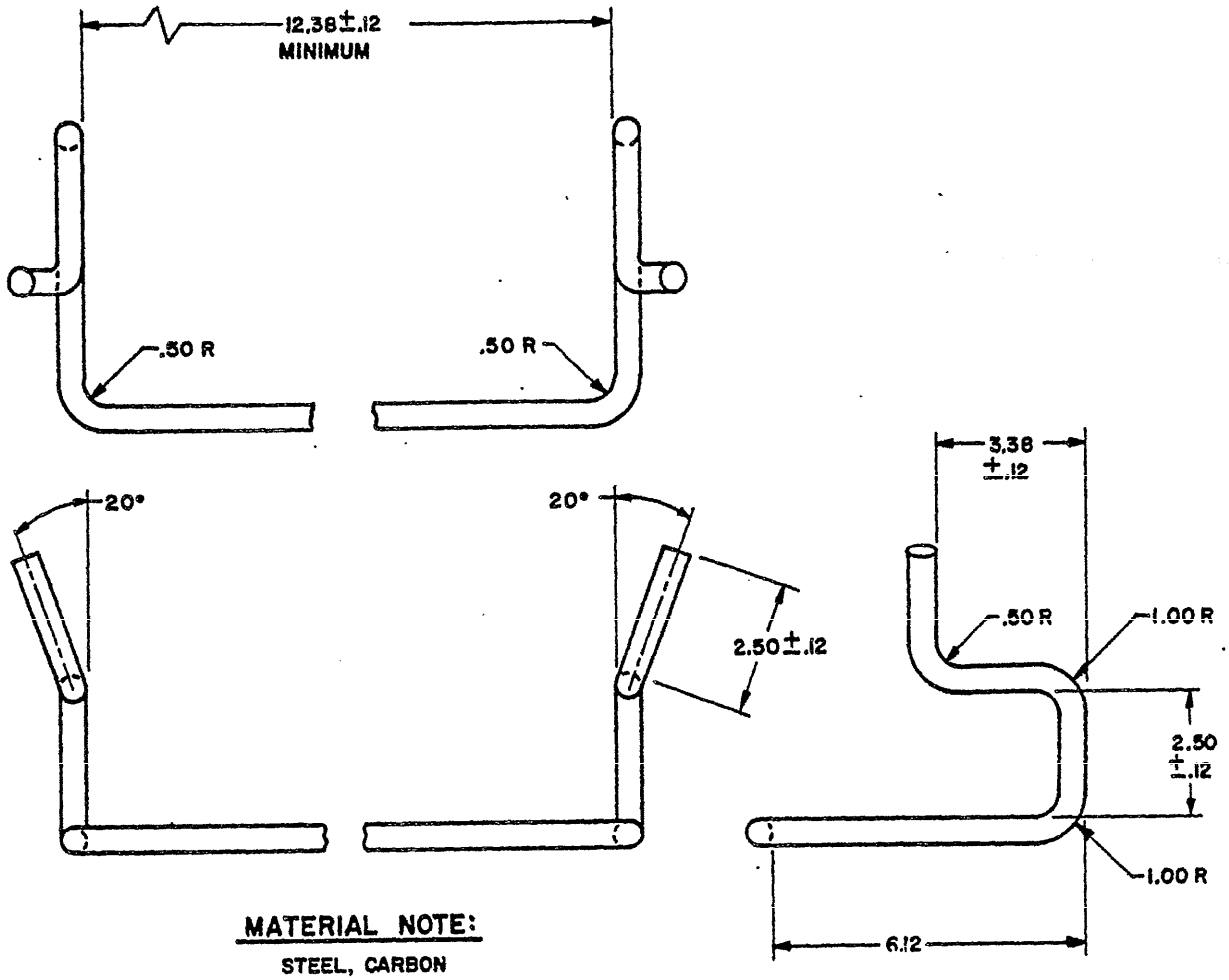
MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 3. CHOCK BLOCK ASSEMBLY

DX-603

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MATERIAL NOTE:

STEEL, CARBON
SAE 1015 TO SAE 1025
1/2 DIA

FIGURE 4. CRADLE

DX-604

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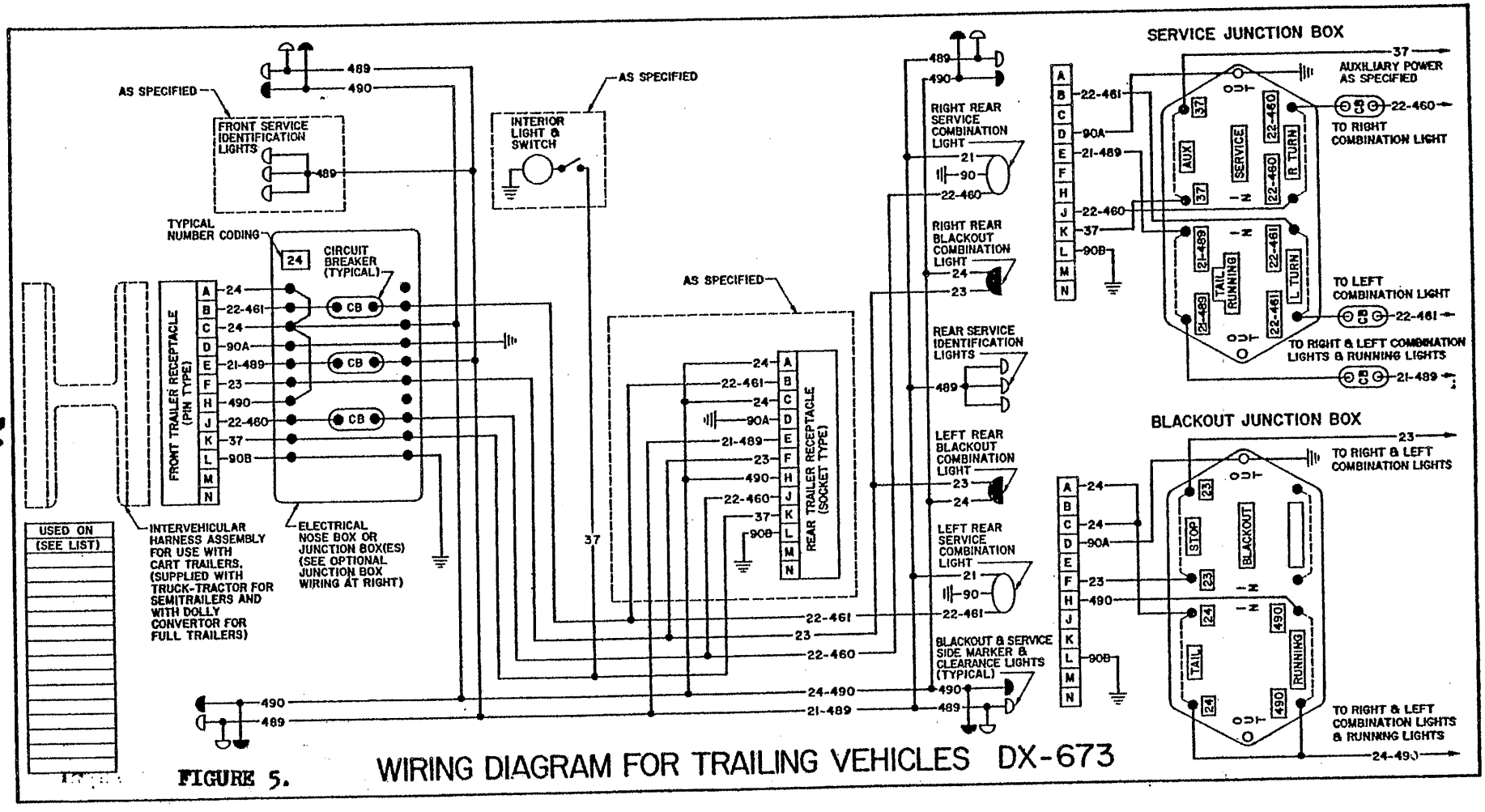


FIGURE 5. WIRING DIAGRAM FOR TRAILING VEHICLES DX-673

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